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Bachelor Thesis

Distributed Algorithms on Complex Networks

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1 Introduction

In this thesis, distributed algorithms for aggregative games on communication networks are studied. A classical example of an aggregative game is the Nash–Cournot game [1]. Many methods for computing a Nash equilibrium assume that players know the aggregate of all decisions. In practice, this assumption is often unrealistic. Each player may only be able to communicate with its neighbors. This motivates the use of distributed algorithms in which each player updates through local communication with neighbors [2].

The communication network matters because it affects how fast information spreads and, as a result, how fast the algorithms converge. Real communication networks are often neither fully regular nor fully random [3]. The Watts–Strogatz model captures this intermediate case, and this thesis focuses on Watts–Strogatz small-world networks to study how such networks affect the convergence of two distributed algorithms.

This thesis builds on research on distributed averaging and gossip. Boyd et al. [4, Sec. II-B] show that the spectral quantity $\lambda_2(\mathbb{E}[W])$ controls the convergence rate of randomized gossip. Koshal et al. [2] bring these gossip ideas to aggregative games, which is the framework reproduced and extended here. More recent work, such as Zhu et al. [5], studies fully asynchronous games over directed networks.

The main focuses of this thesis is numerical experiments. The benchmark results of [2] are reproduced, and the numerical study is extended to Watts–Strogatz small-world networks. In particular, the rewiring probability p_{rew} is examined to see how it affects the convergence of the two distributed algorithms.

The thesis is organized as follows. Chapter 2 introduces the basic definitions and the graph background. Chapter 3 formulates the aggregative game as a variational inequality and presents the Nash–Cournot model. Chapter 4 introduces the two distributed algorithms. Chapter 5 presents their convergence results and provides the theoretical basis for the experiments. Chapter 6 reports the numerical experiments. Chapter 7 concludes and discusses limitations and possible directions for future work.

2 Foundations

This chapter reviews the main background used in the thesis. Unless stated otherwise, Sections 2.2 and 2.3 are mainly based on [6, Chapter 1–3], [1, p. 66–92], [7, Chapter 1].

2.1 Notation

Throughout this thesis, $\|\cdot\|$ denotes the Euclidean norm on $\mathbb{R}^n$ and $\langle \cdot, \cdot \rangle$ denotes the standard inner product. All vectors are written as column vectors. For a nonempty, closed, and convex set $X \subseteq \mathbb{R}^n$, the Euclidean projection of a point y onto X is denoted by $\Pi_X(y)$. If $x = (x_1, \dots, x_N)$ with $x_i \in \mathbb{R}^n$, then $\|x\|^2 = \sum_{i=1}^N \|x_i\|^2$. The Minkowski sum of two sets X and Y is given by $X + Y = \{x + y \mid x \in X, y \in Y\}$. We write $\mathbb{P}$ for probability and $\mathbb{E}$ for expectation. A random sequence X_k converges almost surely to X if $\mathbb{P}(\lim_{k \rightarrow \infty} X_k = X) = 1$. For a σ -algebra $\mathcal{F}$, $\mathbb{E}[X \mid \mathcal{F}]$ denotes the conditional expectation of X given $\mathcal{F}$. The identity matrix is denoted by $\mathbf{I}$, the all-ones vector by $\mathbf{1}$. The dimensions of $\mathbf{I}$ and $\mathbf{1}$ follow from context.

2.2 Convexity and projection

Definition 2.1. A set $X \subseteq \mathbb{R}^n$ is called convex if for all $x, y \in X$ and all $\lambda \in (0, 1)$, the point $\lambda x + (1 - \lambda)y$ also belongs to X .

Definition 2.2. Let $X \subseteq \mathbb{R}^n$ be a convex set. A function $f : X \rightarrow \mathbb{R}$ is called convex if for all $x, y \in X$ and $\lambda \in (0, 1)$:

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y).$$

Proposition 2.1. (Projection onto a convex set) Let $X \subseteq \mathbb{R}^n$ be nonempty, closed, and convex and let $y \in \mathbb{R}^n$. A point $z \in X$ satisfies $z = \Pi_X(y)$ if and only if:

$$(z - y)^\top (x - z) \geq 0 \quad \text{for all } x \in X.$$

This inequality will be used repeatedly later on. Geometrically, it means that the angle between $y - z$ and $x - z$ is at least 90° for every $x \in X$.

2.3 Aggregative games and variational inequality formulation

Consider a game with players $\mathcal{N} = \{1, \dots, N\}$. Each player $i \in \mathcal{N}$ chooses a strategy x_i from a nonempty set $X_i \subseteq \mathbb{R}^n$ and has an objective function f_i . We write the joint

strategy set and vector as

$$\mathcal{X} = \prod_{i=1}^N X_i, \quad x = (x_1, \dots, x_N) \in \mathcal{X}.$$

Definition 2.3. A point $x^* = (x_1^*, \dots, x_N^*) \in \mathcal{X}$ is a Nash equilibrium if, for each player $i \in \mathcal{N}$,

$$f_i(x_i^*, x_{-i}^*) \leq f_i(x_i, x_{-i}^*) \quad \forall x_i \in X_i$$

The next definition introduces aggregative games in the form used in this thesis [1, Definition 2].

Definition 2.4. A game is called aggregative if, for each player i , there exists a function $f_i : X_i \times \bar{X} \rightarrow \mathbb{R}$ such that

$$\hat{f}_i(x) = f_i(x_i, \bar{x}), \quad i = 1, \dots, N,$$

where $\bar{x} = \sum_{j=1}^N x_j$ is the aggregate of all players' decisions and

$$\bar{X} = \sum_{j=1}^N X_j = \left\{ \sum_{j=1}^N x_j : x_j \in X_j, j = 1, \dots, N \right\} \subseteq \mathbb{R}^n \quad (1)$$

is the Minkowski sum of the strategy sets.

Remark 2.1. In the aggregative game, each player's objective function depends only on two components: player i 's own decision and the aggregate of all player decisions. If $(x_1^*, \dots, x_N^*)$ is a Nash equilibrium, then $\bar{x}^* = \sum_{j=1}^N x_j^*$ is the corresponding equilibrium aggregate.

Example 2.1. (Cournot Duopoly) A classical example of an aggregative game is a Cournot duopoly. Two firms choose production $x_1 \geq 0$ and $x_2 \geq 0$. They sell their goods in a market where the price depends on the total production: $\bar{x} = x_1 + x_2$. Let $p(\bar{x})$ be the market price and let $c_i(x_i)$ be the cost of firm i . Then the profit of firm i is:

$$\pi_i(x_i, \bar{x}) = x_i p(\bar{x}) - c_i(x_i).$$

This is an aggregative game because each firm's profit depends only on its own decision x_i and on the aggregate $\bar{x}$.

A Nash equilibrium of an aggregative game can be characterized by a variational inequality. See [8] for more theory on Nash equilibrium problems. We now introduce this formulation.

Definition 2.5. Let $X \subseteq \mathbb{R}^n$ be a nonempty, closed, and convex set, and let $F : X \rightarrow \mathbb{R}^n$ be a continuous function. A point $x^* \in X$ is called a solution of the variational inequality problem $\text{VI}(X, F)$ if:

$$\langle F(x^*), y - x^* \rangle \geq 0, \quad \forall y \in X. \quad (2)$$

Geometrically, this means that at a solution point x^* , the vector $F(x^*)$ forms a non-obtuse angle with every feasible direction $y - x^*$, so there is no feasible first-order improvement direction.

Using Proposition 2.1, we obtain the following equivalent fixed-point characterization of $\text{VI}(X, F)$.

Proposition 2.2. (Fixed-Point Formulation of $\text{VI}(X, F)$) For $\gamma > 0$, a point $x^* \in X$ solves $\text{VI}(X, F)$ if and only if

$$x^* = \Pi_X[x^* - \gamma F(x^*)] \quad (3)$$

Proof. Let $u = x^* - \gamma F(x^*)$. By Proposition 2.1, x^* satisfies $x^* = \Pi_X(u)$ if and only if $(x^* - u)^\top (y - x^*) \geq 0$ for all $y \in X$. Since $x^* - u = \gamma F(x^*)$ and $\gamma > 0$, this is equivalent to $\langle F(x^*), y - x^* \rangle \geq 0$ for all $y \in X$, i.e., x^* solving $\text{VI}(X, F)$. $\square$

2.4 Graph and matrix preliminaries

Next, we introduce the graph and matrix concepts needed for the distributed algorithms. The following definitions are adapted from Naderi and Pankrashkin [9, Chapter 1], Koshal et al. [2], Mesbahi and Egerstedt [10].

Definition 2.6. An undirected graph is a pair $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = \{1, \dots, N\}$ is a finite set of vertices and $\mathcal{E}$ is the set of edges. For $i, j \in \mathcal{V}$, we write $i \sim j$ if $\{i, j\} \in \mathcal{E}$.

- The neighbor set of $i \in \mathcal{V}$ is $N_i = \{j \in \mathcal{V} : \{i, j\} \in \mathcal{E}\}$, and its degree is $d_i = |N_i|$, i.e., the number of its neighbors.
- A path from i to j is a sequence of distinct vertices $(i_0, i_1, \dots, i_m)$ with $i_0 = i$, $i_m = j$ such that $\{i_{r-1}, i_r\} \in \mathcal{E}$ for all $r = 1, \dots, m$.
- The graph $\mathcal{G}$ is connected if every pair of distinct vertices is linked by a path.

Remark 2.2. In this thesis, we use the graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ to represent the communication network of the game. In standard network science, vertices represent the players and edges represent communication links [11]. Thus, $\{i, j\} \in \mathcal{E}$ means that players i and j can exchange information directly.

Definition 2.7. Let $A = (a_{ij}) \in \mathbb{R}^{N \times N}$ be a nonnegative matrix. A is called row-stochastic if $\sum_{j=1}^N a_{ij} = 1$ for all i , and A is called column-stochastic if $\sum_{i=1}^N a_{ij} = 1$ for all j . A is called doubly stochastic if both properties hold.

Definition 2.8. Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a communication graph. A matrix $W = (w_{ij}) \in \mathbb{R}^{N \times N}$ is called a weight matrix associated with $\mathcal{G}$ if, for all $i \in \mathcal{V}$

$$w_{ij} > 0 \text{ only if } j = i \text{ or } \{i, j\} \in \mathcal{E}$$

In other words, player i assigns positive weight only to its own value and to the values of its neighbors.

2.5 Communication graphs

The numerical experiments used several communication graphs. The four baseline graphs follow the setup in [2]. We then introduce the Watts–Strogatz model [3], which is used to generate the small-world networks studied in Chapter 6.

Cycle The cycle connects each vertex to its two neighbors along a ring. In this graph, each player communicates only with nearby players.

Wheel The Wheel has one central vertex connected to all other vertices. This graph represents a communication pattern with one highly connected hub.

Grid In the numerical experiments, the grid has five columns and $N/5$ rows. Corner vertices have two neighbors, edge vertices three, and interior vertices four. It is a graph with local communication only.

Complete graph The complete graph connects every pair of distinct vertices. In this graph, every player can communicate directly with all other players.

The four baseline graphs are illustrated in Figure 1.

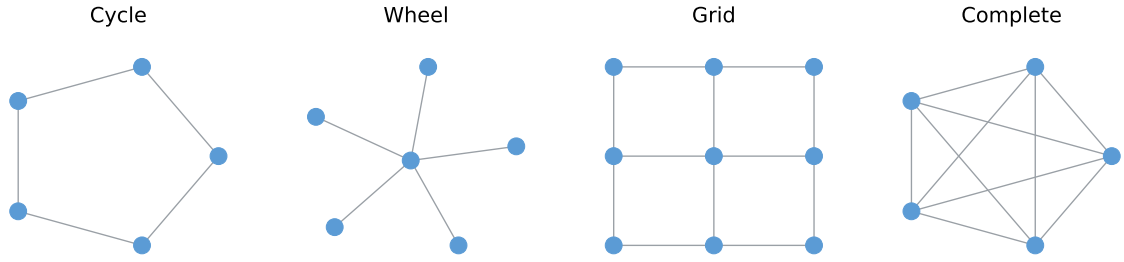


Figure 1: The four baseline graphs.

Source: Generated using Python (NetworkX and Matplotlib).

Small-world network (Watts–Strogatz model) To model networks that are neither fully regular nor fully random, we use the Watts–Strogatz model [3]. This model lies between a regular ring lattice and a random graph through two parameters: the initial degree K and the rewiring probability p_{rew} .

Definition 2.9. The Watts–Strogatz graph $\text{WS}(N, K, p_{\text{rew}})$ is constructed as follows:

- (i) Start with a ring lattice with N vertices, where each vertex is connected to its K nearest neighbors, with $K/2$ neighbors on each side.
- (ii) For each edge from vertex i to its t -th clockwise neighbor, where $t = 1, \dots, K/2$, rewire the edge with probability p_{rew} .

If rewiring takes place, the endpoint is replaced by a new vertex chosen uniformly at random, excluding i and any vertex already adjacent to i . Figure 2 shows how the rewiring step changes the graph.

Definition 2.10. Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a connected undirected graph.

- (i) The characteristic path length $L(\mathcal{G})$ is the average length of the shortest paths between all pairs of vertices in $\mathcal{G}$.
- (ii) The clustering coefficient $C(\mathcal{G})$ measures how strongly the neighbors of a vertex are connected together. More precisely, for each vertex, one considers the fraction of possible edges between its neighbors that actually exist, and $C(\mathcal{G})$ is the average of this quantity over all vertices.

These two quantities are standard structural indicators used to characterize the small-world effect. Intuitively, $L(\mathcal{G})$ measures the global separation of the network, while $C(\mathcal{G})$ measures how strongly neighboring vertices are connected to each other.

Remark 2.3. A graph $\mathcal{G}$ is said to have the small-world effect, in the sense of Watts and Strogatz, if its $C(\mathcal{G})$ is much larger than that of a comparable random graph, while its $L(\mathcal{G})$ remains close to the random-graph value,

$$C(\mathcal{G}) \gg C_{\text{random}}, \quad L(\mathcal{G}) \gtrsim L_{\text{random}}.$$

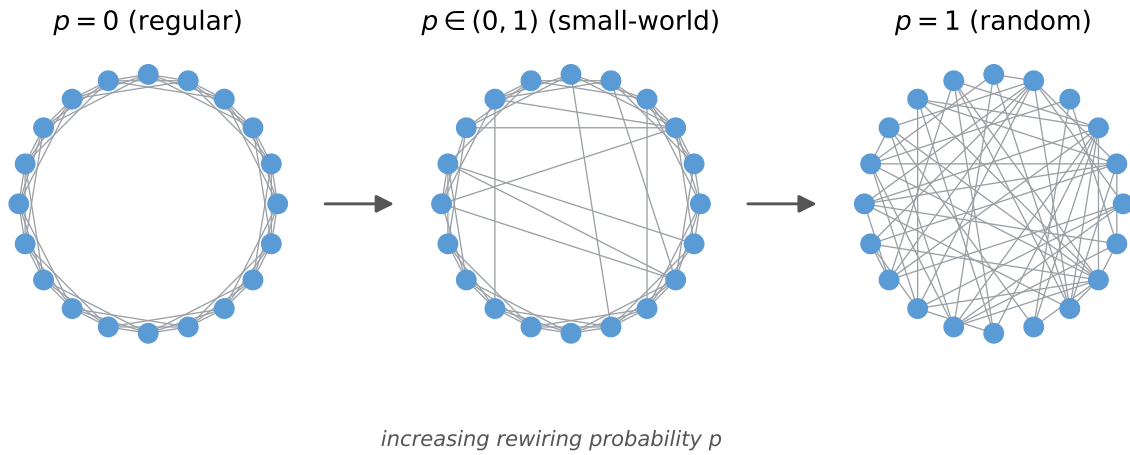


Figure 2: The Watts–Strogatz rewiring procedure for $N = 20$ and $K = 6$. Figure adapted from [3].

Source: Generated using Python (NetworkX and Matplotlib).

3 Model formulation

This chapter formulates the aggregative game as a variational inequality problem and then introduces the Nash–Cournot model used in the numerical experiments.

3.1 Variational inequality and standing assumptions

The following assumptions are standard in the literature on aggregative games and variational inequalities [2, 5].

Assumption 1. For each $i = 1, \dots, N$, the strategy set X_i is compact and convex, and the function $f_i(x_i, u)$ is continuously differentiable in an open set containing $X_i \times \bar{X}$. Moreover, for every fixed $u \in \bar{X}$, the function $f_i(x_i, u)$ is convex in x_i over X_i .

Let

$$\mathcal{X} = \prod_{i=1}^N X_i, \quad x = (x_1^\top, \dots, x_N^\top)^\top \in \mathcal{X}.$$

and let $\bar{X}$ denote the set of feasible aggregate vectors as defined in (1). We use $u \in \bar{X}$ to denote a generic aggregate vector.

For each player $i = 1, \dots, N$, define

$$F_i(x_i, u) = \nabla_{x_i} f_i(x_i, u).$$

Then

$$F(x, u) = \begin{pmatrix} F_1(x_1, u) \\ \vdots \\ F_N(x_N, u) \end{pmatrix}.$$

For each $x \in \mathcal{X}$, let $\bar{x} := \sum_{i=1}^N x_i \in \bar{X}$. The game mapping is then defined by

$$\phi(x) = F(x, \bar{x}).$$

Under Assumption 1, a Nash equilibrium of the game is a solution of the variational inequality $\text{VI}(\mathcal{X}, \phi)$ [8, Proposition 1.4.2]: find $x^* \in \mathcal{X}$ such that

$$\langle \phi(x^*), x - x^* \rangle \geq 0, \quad \forall x \in \mathcal{X}. \quad (4)$$

Assumption 2 (Strict monotonicity) The mapping ϕ is strictly monotone on $\mathcal{X}$; that is,

$$(\phi(x) - \phi(y))^\top (x - y) > 0, \quad \forall x, y \in \mathcal{X}, \ x \neq y.$$

Proposition 3.1 (Existence and uniqueness of Nash equilibrium). Under Assumptions 1 and 2, the aggregative game has a unique Nash equilibrium.

Proof. By Assumption 1, the set $\mathcal{X}$ is compact and convex, and the mapping ϕ is continuous. Therefore, the variational inequality $\text{VI}(\mathcal{X}, \phi)$ has a solution. See [8, Corollary 2.2.6]. By Assumption 2, the mapping ϕ is strictly monotone on $\mathcal{X}$. Therefore, the variational inequality $\text{VI}(\mathcal{X}, \phi)$ has at most one solution, which implies that the Nash equilibrium is unique. See [8, Theorem 2.3.3] for full proofs. $\square$

Assumption 3 (Uniform Lipschitz continuity) For each player i , there exists a constant $L_i > 0$ such that, for every fixed $x_i \in X_i$, the mapping $u \mapsto F_i(x_i, u)$ is Lipschitz continuous on an open set $U \subseteq \mathbb{R}^n$ containing the feasible set $\bar{X}$; that is,

$$\|F_i(x_i, w) - F_i(x_i, z)\| \leq L_i \|w - z\|, \quad \forall w, z \in U.$$

Remark 3.1. Compared with joint Lipschitz continuity assumption used in some related works [5], Assumption 3 is weaker, since it does not require the mapping F_i to be Lipschitz continuous jointly in (x_i, u) over the product set. Instead, it only imposes uniform Lipschitz continuity of $F_i(x_i, \cdot)$ with respect to the second argument.

3.2 Nash–Cournot example and verification of assumptions

The Nash–Cournot game is a classical example of an aggregative game, as already noted in Example 2.1. In this model considered here, the aggregate variable u is the aggregate sales vector $\bar{s}$, and this specialization is used in the numerical experiments. There are N firms competing over $\mathcal{L}$ markets. The market locations are fixed, and transportation costs are ignored.

The model is defined as follows:

- **Players:** There are N firms, indexed by $i = 1, \dots, N$.
- **Decisions:** For each market $\ell = 1, \dots, \mathcal{L}$, firm i chooses a production quantity $g_{i\ell} \geq 0$ and a sales quantity $s_{i\ell} \geq 0$. Define

$$g_i = (g_{i1}, \dots, g_{i\mathcal{L}})^\top, \quad s_i = (s_{i1}, \dots, s_{i\mathcal{L}})^\top,$$

and let

$$x_i = (g_i, s_i)^\top \in \mathbb{R}^{2\mathcal{L}}.$$

- **Aggregate variable:** In this model, the aggregate entering the cost is the aggregate sales vector

$$\bar{s} = (\bar{s}_1, \dots, \bar{s}_{\mathcal{L}})^\top, \quad \bar{s}_\ell = \sum_{i=1}^N s_{i\ell}.$$

- **Inverse demand function:** At market ℓ , the market price is given by $p_\ell(\bar{s}_\ell)$.
- **Production Cost function:** The production cost of firm i at market ℓ is $c_{i\ell}(g_{i\ell})$.
- **Capacity constraint:** The production capacity of firm i at market ℓ is $\text{Cap}_{i\ell} > 0$.
- **Strategy set:** The feasible set of firm i is

$$X_i = \left\{ x_i \in \mathbb{R}_+^{2\mathcal{L}} : \sum_{\ell=1}^{\mathcal{L}} s_{i\ell} \leq \sum_{\ell=1}^{\mathcal{L}} g_{i\ell}, \quad g_{i\ell} \leq \text{Cap}_{i\ell}, \quad \ell = 1, \dots, \mathcal{L} \right\}. \quad (5)$$

- **Profit function:** The profit of firm i is

$$\pi_i(x_i, \bar{s}) = \sum_{\ell=1}^{\mathcal{L}} \left(p_\ell(\bar{s}_\ell) s_{i\ell} - c_{i\ell}(g_{i\ell}) \right).$$

- **Optimization problem:** Each firm solves

$$\max_{x_i \in X_i} \pi_i(x_i, \bar{s}).$$

Equivalently,

$$\min_{x_i \in X_i} f_i(x_i, \bar{s}), \quad f_i(x_i, \bar{s}) = \sum_{\ell=1}^{\mathcal{L}} (c_{i\ell}(g_{i\ell}) - p_\ell(\bar{s}_\ell) s_{i\ell}). \quad (6)$$

We now check that this Nash–Cournot model satisfies Assumptions 1–3. This is needed for the numerical experiments later. To do so, we first write the player mapping $F_i(x_i, u)$ explicitly.

For each firm $i = 1, \dots, N$ and each market $\ell = 1, \dots, \mathcal{L}$, let $x_{i\ell} = (g_{i\ell}, s_{i\ell})^\top$. Then the coordinate map is given by

$$F_{i\ell}(x_{i\ell}, u_\ell) = \begin{pmatrix} c'_{i\ell}(g_{i\ell}) \\ -p_\ell(u_\ell) - p'_\ell(u_\ell) s_{i\ell} \end{pmatrix}. \quad (7)$$

Hence,

$$F_i(x_i, u) = \begin{pmatrix} F_{i1}(x_{i1}, u_1) \\ \vdots \\ F_{i\mathcal{L}}(x_{i\mathcal{L}}, u_{\mathcal{L}}) \end{pmatrix}, \quad F(x, u) = \begin{pmatrix} F_1(x_1, u) \\ \vdots \\ F_N(x_N, u) \end{pmatrix}.$$

For this model, the game mapping is given by

$$\phi(x) = F(x, \bar{s}).$$

We specialize the model to an affine inverse demand function and quadratic production costs, a well-known specification that also fits naturally within broader quadratic-game formulations, see [2, 12].

Let

$$\bar{S} := \left\{ \sum_{i=1}^N s_i \mid x_i = (g_i, s_i) \in X_i, \ i = 1, \dots, N \right\} \subseteq \mathbb{R}^{\mathcal{L}}$$

denote the feasible set of aggregate sales vectors.

Proposition 3.2. (Verification of Assumptions 1–3 for the Nash–Cournot model used in the numerical experiments) Consider the Nash–Cournot game introduced above. For each market $\ell = 1, \dots, \mathcal{L}$, suppose that the affine inverse demand and production cost are given by

$$p_\ell(\bar{s}_\ell) = m_\ell - \bar{s}_\ell, \quad c_{i\ell}(g_{i\ell}) = a_{i\ell}g_{i\ell} + b_{i\ell}g_{i\ell}^2,$$

where $m_\ell > 0$, $a_{i\ell} \geq 0$, $b_{i\ell} > 0$, and $\text{Cap}_{i\ell} > 0$ for all $i \in \mathcal{N}$ and $\ell = 1, \dots, \mathcal{L}$. Then the Nash–Cournot model satisfies Assumptions 1–3. Consequently, the game admits a unique Nash equilibrium.

Proof. We verify Assumptions 1–3.

A1. Since X_i is a bounded polytope, it is compact and convex.

For fixed $\bar{s} \in \bar{S}$, the local objective function is

$$f_i(x_i, \bar{s}) = \sum_{\ell=1}^{\mathcal{L}} \left(a_{i\ell}g_{i\ell} + b_{i\ell}g_{i\ell}^2 - (m_\ell - \bar{s}_\ell)s_{i\ell} \right).$$

This function is continuously differentiable on an open set $X_i \times \bar{S}$. Moreover, for fixed $\bar{s}$, it is convex in x_i , because each term

$$a_{i\ell}g_{i\ell} + b_{i\ell}g_{i\ell}^2$$

is convex in $g_{i\ell}$ since $b_{i\ell} > 0$, while

$$-(m_\ell - \bar{s}_\ell)s_{i\ell}$$

is linear in $s_{i\ell}$. Hence $f_i(\cdot, \bar{s})$ is convex on X_i , and Assumption 1 holds.

A2. Let $x = (x_1, \dots, x_N), y = (y_1, \dots, y_N) \in X$ with $x \neq y$, where

$$x_i = (g_i, s_i), \quad y_i = (h_i, t_i),$$

and write

$$\bar{s}_\ell = \sum_{i=1}^N s_{i\ell}, \quad \bar{t}_\ell = \sum_{i=1}^N t_{i\ell}, \quad \ell = 1, \dots, \mathcal{L}.$$

Under the specialized model,

$$c'_{i\ell}(g_{i\ell}) = a_{i\ell} + 2b_{i\ell}g_{i\ell}, \quad p_\ell(r) = m_\ell - r, \quad p'_\ell(r) = -1.$$

Hence, by (7),

$$F_{i\ell}(x_{i\ell}, \bar{s}_\ell) = \begin{pmatrix} a_{i\ell} + 2b_{i\ell}g_{i\ell} \\ -m_\ell + \bar{s}_\ell + s_{i\ell} \end{pmatrix},$$

and similarly,

$$F_{i\ell}(y_{i\ell}, \bar{t}_\ell) = \begin{pmatrix} a_{i\ell} + 2b_{i\ell}h_{i\ell} \\ -m_\ell + \bar{t}_\ell + t_{i\ell} \end{pmatrix}.$$

Therefore,

$$F_{i\ell}(x_{i\ell}, \bar{s}_\ell) - F_{i\ell}(y_{i\ell}, \bar{t}_\ell) = \begin{pmatrix} 2b_{i\ell}(g_{i\ell} - h_{i\ell}) \\ (\bar{s}_\ell - \bar{t}_\ell) + (s_{i\ell} - t_{i\ell}) \end{pmatrix}.$$

It follows that

$$(\phi(x) - \phi(y))^\top (x - y) = \sum_{i=1}^N \sum_{\ell=1}^{\mathcal{L}} \left[2b_{i\ell}(g_{i\ell} - h_{i\ell})^2 + ((\bar{s}_\ell - \bar{t}_\ell) + (s_{i\ell} - t_{i\ell}))(s_{i\ell} - t_{i\ell}) \right].$$

Summing the second term over i gives

$$\sum_{i=1}^N ((\bar{s}_\ell - \bar{t}_\ell) + (s_{i\ell} - t_{i\ell}))(s_{i\ell} - t_{i\ell}) = (\bar{s}_\ell - \bar{t}_\ell) \sum_{i=1}^N (s_{i\ell} - t_{i\ell}) + \sum_{i=1}^N (s_{i\ell} - t_{i\ell})^2.$$

Since

$$\sum_{i=1}^N (s_{i\ell} - t_{i\ell}) = \bar{s}_\ell - \bar{t}_\ell,$$

we obtain

$$\sum_{i=1}^N ((\bar{s}_\ell - \bar{t}_\ell) + (s_{i\ell} - t_{i\ell}))(s_{i\ell} - t_{i\ell}) = (\bar{s}_\ell - \bar{t}_\ell)^2 + \sum_{i=1}^N (s_{i\ell} - t_{i\ell})^2.$$

Hence

$$(\phi(x) - \phi(y))^\top (x - y) = \sum_{\ell=1}^{\mathcal{L}} \left[\sum_{i=1}^N 2b_{i\ell}(g_{i\ell} - h_{i\ell})^2 + (\bar{s}_\ell - \bar{t}_\ell)^2 + \sum_{i=1}^N (s_{i\ell} - t_{i\ell})^2 \right].$$

Since $b_{i\ell} > 0$ for all i, ℓ , every term on the right-hand side is nonnegative, and because

$x \neq y$. Therefore,

$$(\phi(x) - \phi(y))^\top (x - y) > 0 \quad \text{for all } x \neq y.$$

Thus, ϕ is strictly monotone on X , and Assumption 2 holds.

A3. Fix $x_i \in X_i$ and let $u, z \in \bar{S}$. Under the affine inverse demand

$$p_\ell(r) = m_\ell - r,$$

we have $p'_\ell(r) = -1$ for all $r \in \bar{S}_\ell$. Hence, by (7),

$$F_{i\ell}(x_{i\ell}, u_\ell) = \begin{pmatrix} a_{i\ell} + 2b_{i\ell}g_{i\ell} \\ -m_\ell + u_\ell + s_{i\ell} \end{pmatrix}.$$

Therefore,

$$F_{i\ell}(x_{i\ell}, u_\ell) - F_{i\ell}(x_{i\ell}, z_\ell) = \begin{pmatrix} 0 \\ u_\ell - z_\ell \end{pmatrix}.$$

It follows that

$$\|F_i(x_i, u) - F_i(x_i, z)\|^2 = \sum_{\ell=1}^{\mathcal{L}} |u_\ell - z_\ell|^2 = \|u - z\|^2.$$

Hence

$$\|F_i(x_i, u) - F_i(x_i, z)\| \leq \|u - z\| \quad \text{for all } u, z \in \bar{S},$$

and therefore $F_i(x_i, \cdot)$ is Lipschitz continuous with constant $L_i = 1$. □

4 Distributed algorithms

In the distributed setting, players cannot observe the true aggregate directly. Instead, each player keeps a local estimate and updates it through communication with its neighbors. Following [2], we present two algorithms: a synchronous algorithm and an asynchronous gossip-based algorithm for aggregative games.

4.1 Algorithmic notation

The following notation is used in this chapter and Chapter 5.

$\mathcal{G}_k, \mathcal{G}$	communication graph at iteration k and static communication graph
N_i^k, N_i	neighbor set of player i
$W(k), w_{ij}(k)$	weight matrix and its entries
v_i^k	player i 's local estimate of the average aggregate
$\hat{v}_i^k$	estimate after the mixing step
$N\hat{v}_i^k$	player i 's estimate of the aggregate sum
$\alpha_k, \alpha_{k,i}$	stepsize in the synchronous and asynchronous algorithms
I^k, J^k	pair of players selected at iteration k in the asynchronous algorithms
p_{ij}	probability that player i contacts neighbor j
Γ_i^k	number of updates of player i up to iteration k

4.2 Synchronous distributed algorithm

In the synchronous scheme, all players update at every iteration. We first present the method for time-varying communication graphs. The static graph case used in Chapter 6 is a special case.

Let

$$\mathcal{G}_k = (\mathcal{V}, \mathcal{E}_k),$$

where $\mathcal{V} = \{1, \dots, N\}$ is the set of players and $\mathcal{E}_k$ is the set of communication links at iteration k . For each player i , define

$$N_i^k = \{j \in \mathcal{V} : \{i, j\} \in \mathcal{E}_k\} \cup \{i\}.$$

Thus, N_i^k contains player i and all neighbors of player i at iteration k .

We use the following assumptions on the communication process. They are standard in distributed optimization over time-varying graphs ([13], [2]).

Assumption 4 (Connectivity) There exists an integer $B \geq 1$ such that, for every $k \geq 0$, the union graph

$$(\mathcal{V}, \mathcal{E}_k \cup \mathcal{E}_{k+1} \cup \dots \cup \mathcal{E}_{k+B-1})$$

is connected.

Remark 4.1. When the communication graph is static, that is, $\mathcal{G}_k = \mathcal{G}$ for all $k \geq 0$, then Assumption 4 holds with $B = 1$ exactly when $\mathcal{G}$ is connected. The static case is included in the time-varying setting.

Assumption 5 (Weight matrix) Let $W(k) = (w_{ij}(k)) \in \mathbb{R}^{N \times N}$ be the weight matrix at iteration k . There exists a constant $\delta \in (0, 1)$ such that

$$w_{ij}(k) \geq \delta \quad \text{for all } j \in N_i^k, \quad w_{ij}(k) = 0 \quad \text{for all } j \notin N_i^k,$$

and $W(k)$ is doubly stochastic, that is,

$$\sum_{j=1}^N w_{ij}(k) = 1 \quad \text{for all } i, \quad \sum_{i=1}^N w_{ij}(k) = 1 \quad \text{for all } j.$$

Assumption 6 (Stepsize) The stepsize sequence $\{\alpha_k\}$ is positive, nonincreasing, and satisfies

$$\sum_{k=0}^{\infty} \alpha_k = \infty, \quad \sum_{k=0}^{\infty} \alpha_k^2 < \infty.$$

Under Assumptions 4–6, each player updates its local estimate and decision by the following three steps. The player first averages the estimates received from its neighbors, then takes a projected gradient step using the estimated aggregate, and finally updates its estimate by adding the change in its own decision. This leads to Algorithm 1.

Algorithm 1 Synchronous distributed algorithm

1. Initialize $x_i^0 \in X_i$ and set $v_i^0 = x_i^0$ for all $i = 1, \dots, N$.

2. For $k = 0, 1, 2, \dots$, each player i performs:

a) Mixing step:

$$\hat{v}_i^k = \sum_{j=1}^N w_{ij}(k) v_j^k. \quad (8)$$

b) Decision update:

$$x_i^{k+1} = \Pi_{X_i} [x_i^k - \alpha_k F_i(x_i^k, N\hat{v}_i^k)]. \quad (9)$$

c) Estimate update:

$$v_i^{k+1} = \hat{v}_i^k + x_i^{k+1} - x_i^k. \quad (10)$$

4.3 Asynchronous distributed algorithm

The synchronous scheme requires all players to update at the same time. This is often unrealistic. We now consider an asynchronous gossip-based version. In this setting, at each iteration, only one communicating pair updates.

In this section, the communication network is modeled by a static undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. For each player i , let

$$N_i = \{j \in \mathcal{V} : \{i, j\} \in \mathcal{E}\}$$

denote the set of neighbors of player i .

As in the synchronous case, we impose a connectivity condition on the communication graph.

Assumption 7 The undirected graph $\mathcal{G}$ is connected.

Each player has an independent rate-1 Poisson clock. We use $k = 0, 1, 2, \dots$ to count the total number of clock ticks in the network. At each tick, exactly one player wakes up. We denote this player by I^k . Player I^k then contacts one of its neighbors $J^k \in N_{I^k}$ with probability $p_{ij} > 0$. Only the pair (I^k, J^k) updates at iteration k ; all other players remain unchanged.

The asynchronous distributed algorithm follows the same basic structure as the synchronous one. The selected players first mix their aggregate estimates, then update their decisions by a projected gradient step, and finally update their local estimates. This leads to Algorithm 2.

Algorithm 2 Asynchronous distributed algorithm

1. Initialize $x_i^0 \in X_i$ and set $v_i^0 = x_i^0$ for all $i = 1, \dots, N$.
2. For $k = 0, 1, 2, \dots$, select a communicating pair (I^k, J^k) .
 - a) Mixing step: for all $i = 1, \dots, N$, define

$$\hat{v}_i^k = \sum_{j=1}^N [W(k)]_{ij} v_j^k,$$

where

$$W(k) = \mathbf{I} - \frac{1}{2}(e_{I^k} - e_{J^k})(e_{I^k} - e_{J^k})^\top, \quad (11)$$

and e_i is the i -th unit vector in $\mathbb{R}^N$. The matrix $W(k)$ is doubly stochastic.

In particular,

$$\hat{v}_i^k = \begin{cases} \frac{1}{2}(v_{I^k}^k + v_{J^k}^k), & i \in \{I^k, J^k\}, \\ v_i^k, & i \notin \{I^k, J^k\}. \end{cases}$$

- b) Decision update: for $i \in \{I^k, J^k\}$,

$$x_i^{k+1} = \Pi_{X_i} \left[x_i^k - \alpha_{k,i} F_i(x_i^k, N\hat{v}_i^k) \right]. \quad (12)$$

For $i \notin \{I^k, J^k\}$, let $x_i^{k+1} = x_i^k$.

- c) Estimate update:

$$v_i^{k+1} = \hat{v}_i^k + x_i^{k+1} - x_i^k. \quad (13)$$

Equivalently, for $i \notin \{I^k, J^k\}$ we have $v_i^{k+1} = v_i^k$.

To define the player-dependent stepsize, let Γ_i^k denote the number of updates of player i up to and including iteration k . For selected players, set

$$\alpha_{k,i} = \frac{1}{\Gamma_i^k}, \quad i = 1, \dots, N.$$

Remark 4.2. The matrix $W(k)$ in gossip update (11) describes the local averaging step between the selected pair. It also gives a convenient matrix form of the gossip update, which will be used again in Chapter 5.

5 Convergence theory

This chapter proves convergence of the two algorithms from Chapter 4. Under the standing assumptions, the iterates converge almost surely to the unique Nash equilibrium x^* of $\text{VI}(\mathcal{X}, \phi)$. The convergence analysis follows [2]. The main arguments are rewritten here in the notation of this thesis.

Here we list three standard inequalities used in the proofs, following [13]. The Euclidean norm is convex. For any vectors $x_1, \dots, x_m \in \mathbb{R}^n$ and any nonnegative scalars $\mu_1, \dots, \mu_m$ with $\sum_{i=1}^m \mu_i = 1$, we have

$$\left\| \sum_{i=1}^m \mu_i x_i \right\| \leq \sum_{i=1}^m \mu_i \|x_i\|. \quad (14)$$

The squared norm is also convex. Under the same conditions,

$$\left\| \sum_{i=1}^m \mu_i x_i \right\|^2 \leq \sum_{i=1}^m \mu_i \|x_i\|^2. \quad (15)$$

A special form of (15) is $\|a + b\|^2 \leq 2(\|a\|^2 + \|b\|^2)$ for any $a, b \in \mathbb{R}^n$.

The Euclidean projection onto a nonempty, closed, and convex set $X \subseteq \mathbb{R}^n$ is nonexpansive:

$$\|\Pi_X(a) - \Pi_X(b)\| \leq \|a - b\| \quad \text{for all } a, b \in \mathbb{R}^n. \quad (16)$$

The convergence proofs below also rely on two classical results from stochastic approximation theory.

Lemma 5.1. ([14, Lemma 11]) Let $\{V_k\}, \{a_k\}, \{H_k\}, \{R_k\}$ be nonnegative random variables adapted to a filtration $\{\mathcal{F}_k\}$ such that

$$\mathbb{E}[V_{k+1} \mid \mathcal{F}_k] \leq (1 + a_k)V_k - R_k + H_k \quad \text{a.s.},$$

with $\sum_{k=0}^{\infty} a_k < \infty$ and $\sum_{k=0}^{\infty} H_k < \infty$ a.s. Then V_k converges almost surely, and $\sum_{k=0}^{\infty} R_k < \infty$.

Lemma 5.2. ([13, Lemma 3.1(b)]) Let $\{\zeta_k\}$ be a nonnegative sequence with $\sum_{k=0}^{\infty} \zeta_k < \infty$, and let $\beta \in (0, 1)$. Then $\sum_{k=0}^{\infty} \sum_{s=0}^k \beta^{k-s} \zeta_s < \infty$.

5.1 Convergence of the synchronous algorithm

Before proving Theorem 5.1, four auxiliary results are needed. They describe the mixing behavior of the communication process, the relation between the network average and

the players' decisions, a uniform bound on the player mappings, and a disagreement estimate for the local aggregate estimates.

5.1.1 Preliminary lemmas

The repeated mixing step in the algorithm uses products of weight matrices. Under Assumption 5, each $W(k)$ is doubly stochastic. We define:

$$\Phi(k, s) = W(k)W(k-1) \cdots W(s+1)W(s) \quad \text{for } 0 \leq s \leq k.$$

Let $[\Phi(k, s)]_{ij}$ denote the (i, j) -th entry of $\Phi(k, s)$, and $\mathbf{1} \in \mathbb{R}^N$ be the all-ones vector. The next lemma states two properties of $\Phi(k, s)$: the limit and the geometric bound on the mixing error. The result is adapted from [2, Lemma 1]; see also [15, Lemma 5.2.1].

Lemma 5.3. Let Assumptions 4 and 5 hold. Then

- (a) $\lim_{k \rightarrow \infty} \Phi(k, s) = \frac{1}{N} \mathbf{1} \mathbf{1}^\top$ for every $s \geq 0$.
- (b) There exist constants θ and β such that $\left| [\Phi(k, s)]_{ij} - \frac{1}{N} \right| \leq \theta \beta^{k-s}$ for all $0 \leq s \leq k$,
where $\theta = \left(1 - \frac{\delta}{4N^2}\right)^{-2}$ and $\beta = \left(1 - \frac{\delta}{4N^2}\right)^{1/B}$.

Define the network average of the local estimates as, for all $k \geq 0$

$$y^k = \frac{1}{N} \sum_{i=1}^N v_i^k. \tag{17}$$

Although individual estimates v_i^k may differ from one another, their network average y^k tracks the average of the players' decisions x_i^k exactly. We make this precise in the next lemma. The next lemma is adapted from [2, Lemma 2].

Lemma 5.4. Let $W(k)$ be column-stochastic for every $k \geq 0$, and let $v_i^0 = x_i^0$ for $i = 1, \dots, N$. Then for every $k \geq 0$,

$$y^k = \frac{1}{N} \sum_{i=1}^N x_i^k,$$

where y^k is defined by (17).

Proof. We prove $\sum_i v_i^k = \sum_i x_i^k$ by induction on k .

Base case. For $k = 0$, $v_i^0 = x_i^0$ gives the claim.

Induction step. Assume $\sum_i v_i^k = \sum_i x_i^k$ (induction hypothesis, IH). By (8) and (10),

$$v_i^{k+1} = \sum_{j=1}^N w_{ij}(k) v_j^k + (x_i^{k+1} - x_i^k).$$

Summing over i :

$$\sum_{i=1}^N v_i^{k+1} = \sum_{i=1}^N \sum_{j=1}^N w_{ij}(k) v_j^k + \sum_{i=1}^N (x_i^{k+1} - x_i^k). \quad (18)$$

Because $\sum_i w_{ij}(k) = 1$, equation (18) can be written as

$$\sum_{i=1}^N v_i^{k+1} = \sum_{j=1}^N v_j^k + \sum_{i=1}^N (x_i^{k+1} - x_i^k) \stackrel{(\text{IH})}{=} \sum_{j=1}^N x_j^k + \sum_{i=1}^N (x_i^{k+1} - x_i^k) = \sum_{i=1}^N x_i^{k+1}.$$

Dividing by N gives the claim. $\square$

The next lemma provides a uniform bound on the player mappings evaluated at both the true aggregate and the estimated aggregate. This result will be used later in the convergence proof. The next lemma is adapted from [2, Lemma 3].

Lemma 5.5. Let Assumptions 1 and 3 hold, and let $W(k)$ be column-stochastic for every $k \geq 0$. Then there exists a constant $C > 0$ such that for all i and $k \geq 0$

$$\|F_i(x_i^k, Ny^k)\| \leq C, \quad \|F_i(x_i^k, N\hat{v}_i^k)\| \leq C.$$

Proof. We first bound $F_i(x_i^k, Ny^k)$. By Lemma 5.4,

$$Ny^k = \sum_{j=1}^N x_j^k \in \bar{X} \quad \text{for all } k \geq 0.$$

Moreover, $x_i^k \in X_i$ for all $k \geq 0$. Hence,

$$(x_i^k, Ny^k) \in X_i \times \bar{X} \quad \text{for all } k \geq 0.$$

By Assumption 1, F_i is continuous on $X_i \times \bar{X}$. Since both X_i and $\bar{X}$ are compact, the product set $X_i \times \bar{X}$ is compact. Therefore,

$$C_{1,i} = \sup_{(x_i, u) \in X_i \times \bar{X}} \|F_i(x_i, u)\| < \infty.$$

It follows that

$$\|F_i(x_i^k, Ny^k)\| \leq C_{1,i} \quad \text{for all } k \geq 0.$$

Define $C_1 = \max_{1 \leq i \leq N} C_{1,i}$, and therefore

$$\|F_i(x_i^k, Ny^k)\| \leq C_1 \quad \text{for all } i \text{ and } k \geq 0.$$

We next bound $F_i(x_i^k, N\hat{v}_i^k)$. Let

$$Q = \text{conv}\left(\bigcup_{j=1}^N X_j\right),$$

where Q denotes the convex hull of $\bigcup_{j=1}^N X_j$. Since X_j is compact, so is Q . By Lemma 5.4, y^k is a convex combination of points from $\bigcup_{j=1}^N X_j$. Therefore, $y^k \in Q$ for all $k \geq 0$.

By iterating the mixing (8) and estimate update equation (10), we obtain

$$\hat{v}_i^k = \sum_{j=1}^N [\Phi(k, 0)]_{ij} v_j^0 + \sum_{s=1}^k \sum_{j=1}^N [\Phi(k, s)]_{ij} (x_j^s - x_j^{s-1}). \quad (19)$$

While

$$y^k = \frac{1}{N} \sum_{j=1}^N v_j^0 + \frac{1}{N} \sum_{s=1}^k \sum_{j=1}^N (x_j^s - x_j^{s-1}). \quad (20)$$

Subtracting (20) from (19) and applying the triangle inequality, we obtain

$$\|y^k - \hat{v}_i^k\| \leq \sum_{j=1}^N \left| \frac{1}{N} - [\Phi(k, 0)]_{ij} \right| \|v_j^0\| + \sum_{s=1}^k \sum_{j=1}^N \left| \frac{1}{N} - [\Phi(k, s)]_{ij} \right| \|x_j^s - x_j^{s-1}\|.$$

For each j , define

$$D_j = \max_{a, b \in X_j} \|a - b\|.$$

D_j is finite because X_j is compact. Since $x_j^s, x_j^{s-1} \in X_j$, we have $\|x_j^s - x_j^{s-1}\| \leq D_j$. Using Lemma 5.3, we obtain

$$\|y^k - \hat{v}_i^k\| \leq \theta \beta^k \sum_{j=1}^N \|v_j^0\| + \theta \sum_{s=1}^k \beta^{k-s} \sum_{j=1}^N D_j \leq \hat{R}$$

for some constant $\hat{R} > 0$ independent of i and k . Since $y^k \in Q$, we have

$$\hat{v}_i^k \in Q + \text{Ball}(0, \hat{R}) \quad \text{for all } i = 1, k \geq 0.$$

By Assumption 3, the mapping $u \rightarrow F_i(x_i, u)$ is Lipschitz continuous on an open set U containing $N(Q + \text{Ball}(0, \hat{R}))$. Therefore,

$$\|F_i(x_i^k, N\hat{v}_i^k)\| \leq \|F_i(x_i^k, N\hat{v}_i^k) - F_i(x_i^k, Ny^k)\| + \|F_i(x_i^k, Ny^k)\|$$

and hence

$$\|F_i(x_i^k, N\hat{v}_i^k)\| \leq L_i N \|\hat{v}_i^k - y^k\| + C_1 \leq L_i N \hat{R} + C_1.$$

Let

$$L = \max_{1 \leq i \leq N} L_i.$$

Then

$$\|F_i(x_i^k, N\hat{v}_i^k)\| \leq LN\hat{R} + C_1 \quad \text{for all } i \text{ and } k \geq 0.$$

Therefore, with

$$C = \max\{C_1, C_1 + LN\hat{R}\},$$

both estimates hold for all i and $k \geq 0$. $\square$

Lemma 5.6. Let Assumptions 1–6 hold, and let y^k be defined by (17). Then for all $i = 1, \dots, N$ and $k \geq 1$,

$$\|y^k - \hat{v}_i^k\| \leq \theta\beta^k M + \theta NC \sum_{s=1}^k \beta^{k-s} \alpha_{s-1},$$

where θ, β are from Lemma 5.3, C is from Lemma 5.5, and $M = \sum_j \max_{x_j \in X_j} \|x_j\|$.

Proof. Subtracting (20) from (19) and by the triangle inequality, we obtain

$$\|y^k - \hat{v}_i^k\| \leq \sum_{j=1}^N \left| \frac{1}{N} - [\Phi(k, 0)]_{ij} \right| \|v_j^0\| + \sum_{s=1}^k \sum_{j=1}^N \left| \frac{1}{N} - [\Phi(k, s)]_{ij} \right| \|x_j^s - x_j^{s-1}\|.$$

Moreover, by the decision update (9),

$$x_j^s = \Pi_{X_j}(x_j^{s-1} - \alpha_{s-1} F_j(x_j^{s-1}, N\hat{v}_j^{s-1})).$$

Since $x_j^{s-1} \in X_j$, we also have $\Pi_{X_j}(x_j^{s-1}) = x_j^{s-1}$. Hence, by the nonexpansiveness of (16),

$$\begin{aligned} \|x_j^s - x_j^{s-1}\| &= \left\| \Pi_{X_j}(x_j^{s-1} - \alpha_{s-1} F_j(x_j^{s-1}, N\hat{v}_j^{s-1})) - \Pi_{X_j}(x_j^{s-1}) \right\| \\ &\leq \left\| (x_j^{s-1} - \alpha_{s-1} F_j(x_j^{s-1}, N\hat{v}_j^{s-1})) - x_j^{s-1} \right\| \\ &= \alpha_{s-1} \|F_j(x_j^{s-1}, N\hat{v}_j^{s-1})\| \leq \alpha_{s-1} C, \end{aligned}$$

where the last inequality follows from Lemma 5.5.

Using $v_j^0 = x_j^0$, Lemma 5.3 and $M = \sum_{j=1}^N \max_{x_j \in X_j} \|x_j\|$, we obtain

$$\begin{aligned} \|y^k - \hat{v}_i^k\| &\leq \theta \beta^k \sum_{j=1}^N \|v_j^0\| + \theta \sum_{s=1}^k \beta^{k-s} \sum_{j=1}^N \|x_j^s - x_j^{s-1}\| \\ &\leq \theta \beta^k \sum_{j=1}^N \|x_j^0\| + \theta \sum_{s=1}^k \beta^{k-s} \sum_{j=1}^N \alpha_{s-1} C \\ &\leq \theta \beta^k M + \theta N C \sum_{s=1}^k \beta^{k-s} \alpha_{s-1}, \end{aligned}$$

which proves the claim. $\square$

5.1.2 Synchronous convergence theorem

With the previous lemmas, we are now ready to state the convergence result for the synchronous distributed algorithm. Let $x^k = (x_1^k, \dots, x_N^k)$ be the iterates generated by the algorithm. Recall that x^* is the unique solution of $\text{VI}(\mathcal{X}, \phi)$.

Theorem 5.1 (See [2, Prop. 2]). Let Assumptions 1–6 hold. Then the sequence $\{x^k\}$ generated by the synchronous distributed algorithm (8)–(10) converges almost surely to the unique solution x^* of $\text{VI}(\mathcal{X}, \phi)$, i.e., the unique Nash equilibrium:

$$x^k \xrightarrow{\text{a.s.}} x^* \quad \text{as } k \rightarrow \infty.$$

Proof. Let

$$V_k = \|x^k - x^*\|^2. \tag{21}$$

We show $V_k \rightarrow 0$ via Lemma 5.1.

Step 1. Since x^* solves $\text{VI}(\mathcal{X}, \phi)$, the fixed-point formulation (3) gives $x_i^* = \Pi_{X_i}[x_i^* - \alpha_k F_i(x_i^*, \bar{x}^*)]$ for any $\alpha_k > 0$. Using the nonexpansiveness of Π_{X_i} (16) and the update rule (9), we obtain:

$$\begin{aligned} \|x_i^{k+1} - x_i^*\|^2 &\leq \left\| (x_i^k - x_i^*) - \alpha_k (F_i(x_i^k, N\hat{v}_i^k) - F_i(x_i^*, \bar{x}^*)) \right\|^2 \\ &= \|x_i^k - x_i^*\|^2 - 2\alpha_k T_i^k + \alpha_k^2 \|F_i(x_i^k, N\hat{v}_i^k) - F_i(x_i^*, \bar{x}^*)\|^2, \end{aligned}$$

where

$$T_i^k = \left(F_i(x_i^k, N\hat{v}_i^k) - F_i(x_i^*, \bar{x}^*) \right)^\top (x_i^k - x_i^*).$$

For the squared term, using the special case of (15), we obtain:

$$\|F_i(x_i^k, N\hat{v}_i^k) - F_i(x_i^*, \bar{x}^*)\|^2 \leq 2\|F_i(x_i^k, N\hat{v}_i^k)\|^2 + 2\|F_i(x_i^*, \bar{x}^*)\|^2 \leq 2C^2 + 2C^2,$$

By Lemma 5.5, the right-hand side is uniformly bounded, so there exists a constant $\tilde{C} = 4C^2$ such that

$$\sum_{i=1}^N \|F_i(x_i^k, N\hat{v}_i^k) - F_i(x_i^*, \bar{x}^*)\|^2 \leq \tilde{C} \quad \text{for all } k.$$

Step 2. By Lemma 5.4, we have $Ny^k = \sum_j x_j^k$. Starting from the term T_i^k defined in Step 1, we add and subtract $F_i(x_i^k, Ny^k)$ to obtain:

$$T_i^k = T_{i,1}^k + T_{i,2}^k,$$

where

$$\begin{aligned} T_{i,1}^k &= (F_i(x_i^k, Ny^k) - F_i(x_i^*, \bar{x}^*))^\top (x_i^k - x_i^*), \\ T_{i,2}^k &= (F_i(x_i^k, N\hat{v}_i^k) - F_i(x_i^k, Ny^k))^\top (x_i^k - x_i^*). \end{aligned}$$

Step 3. Since $\phi(x) = F(x, \bar{x})$ and $Ny^k = \bar{x}$, it follows

$$\sum_{i=1}^N T_{i,1}^k = \sum_{i=1}^N \left(F_i(x_i^k, Ny^k) - F_i(x_i^*, \bar{x}^*) \right)^\top (x_i^k - x_i^*) = (\phi(x^k) - \phi(x^*))^\top (x^k - x^*).$$

By Assumption 2, the mapping ϕ is strictly monotone on $\mathcal{X}$. Therefore,

$$(\phi(x^k) - \phi(x^*))^\top (x^k - x^*) \geq 0,$$

follows

$$\sum_{i=1}^N T_{i,1}^k \geq 0,$$

with equality only when $x^k = x^*$.

Step 4. Next, we estimate the error term $T_{i,2}^k$. By Cauchy–Schwarz inequality i.e., $\langle a, b \rangle \geq -\|a\| \|b\|$ and Lipschitz continuity of F_i in its second argument (Assumption 3),

$$T_{i,2}^k = \left(F_i(x_i^k, N\hat{v}_i^k) - F_i(x_i^k, Ny^k) \right)^\top (x_i^k - x_i^*) \geq -\|F_i(x_i^k, N\hat{v}_i^k) - F_i(x_i^k, Ny^k)\| \|x_i^k - x_i^*\|$$

and

$$\|F_i(x_i^k, N\hat{v}_i^k) - F_i(x_i^k, Ny^k)\| \leq L_i \|N\hat{v}_i^k - Ny^k\| = L_i N \|\hat{v}_i^k - y^k\|.$$

Therefore,

$$T_{i,2}^k \geq -L_i N \|\hat{v}_i^k - y^k\| \|x_i^k - x_i^*\|.$$

Since $\mathcal{X}$ is compact by Assumption 1, there exists a constant $\hat{C} > 0$ such that

$$\|x_i^k - x_i^*\| \leq \hat{C} \quad \text{for all } i \text{ and } k.$$

Hence,

$$T_{i,2}^k \geq -L_i N \hat{C} \|\hat{v}_i^k - y^k\|.$$

Summing over i , we obtain

$$\sum_{i=1}^N T_{i,2}^k \geq -N \hat{C} \sum_{i=1}^N L_i \|\hat{v}_i^k - y^k\|.$$

and letting $\bar{L} = \sum_{i=1}^N L_i$, By Lemma 5.6,

$$\sum_{i=1}^N T_{i,2}^k \geq -\bar{L} N \hat{C} \left(\theta \beta^k M + \theta N C \sum_{s=1}^k \beta^{k-s} \alpha_{s-1} \right).$$

Step 5. To apply 5.1 we combine Steps 1–4, and write (21) the recursive inequality form

$$V_{k+1} \leq V_k - R_k + H_k, \tag{22}$$

where $R_k = 2\alpha_k \sum_{i=1}^N T_{i,1}^k \geq 0$ and

$$H_k = \underbrace{2\alpha_k \bar{L} N \hat{C} \theta \beta^k M}_{(1)} + \underbrace{2\alpha_k \bar{L} N \hat{C} \theta N C \sum_{s=1}^k \beta^{k-s} \alpha_{s-1}}_{(2)} + \underbrace{\alpha_k^2 \tilde{C}}_{(3)}.$$

Step 6. We show $\sum_k H_k < \infty$. By Assumption 6, term (3)

$$\sum_{k=0}^{\infty} \alpha_k^2 \tilde{C} < \infty.$$

By term (1) we apply Cauchy–Schwarz, because $\sum_k \alpha_k^2 < \infty$ and $\beta < 1$, give

$$\sum_{k=0}^{\infty} \alpha_k \beta^k \leq \left(\sum_{k=0}^{\infty} \alpha_k^2 \right)^{1/2} \left(\sum_{k=0}^{\infty} \beta^{2k} \right)^{1/2} < \infty.$$

By Lemma 5.2 with $\zeta_k = \alpha_k$, after an index shift term (2)

$$\sum_{k=0}^{\infty} \alpha_k \sum_{s=1}^k \beta^{k-s} \alpha_{s-1} < \infty.$$

Therefore,

$$\sum_{k=0}^{\infty} H_k < \infty.$$

Step 7. Applying Lemma 5.1 to (22) with $a_k = 0$, we obtain that, almost surely, V_k converges and

$$R_k = \sum_{k=0}^{\infty} \alpha_k \sum_{i=1}^N T_{i,1}^k < \infty.$$

Since $\sum_{k=0}^{\infty} \alpha_k = \infty$ by Assumption 6, there exists a subsequence $\{k_\ell\}$ such that

$$\sum_{i=1}^N T_{i,1}^{k_\ell} \rightarrow 0.$$

Since $\mathcal{X}$ is compact, by the Bolzano–Weierstrass theorem, there exists a subsequence $\{x^{k_\ell_j}\}$ converging to some $\bar{x} \in \mathcal{X}$. By continuity of ϕ ,

$$\sum_{i=1}^N T_{i,1}^{k_\ell_j} = \left(\phi(x^{k_\ell_j}) - \phi(x^*) \right)^\top (x^{k_\ell_j} - x^*) \rightarrow \left(\phi(\bar{x}) - \phi(x^*) \right)^\top (\bar{x} - x^*).$$

Hence

$$\left(\phi(\bar{x}) - \phi(x^*) \right)^\top (\bar{x} - x^*) = 0.$$

By strict monotonicity of ϕ , this implies $\bar{x} = x^*$. Therefore,

$$V_{k_\ell_j} = \|x^{k_\ell_j} - x^*\|^2 \rightarrow 0.$$

Since $V_k \geq 0$, it follows that $\liminf_{k \rightarrow \infty} V_k = 0$. Because V_k converges almost surely, we conclude that $V_k \rightarrow 0$ a.s. Hence, $x^k \rightarrow x^*$ a.s. $\square$

5.2 Convergence of the asynchronous algorithm

We now turn to the asynchronous gossip algorithm from Section 4.3. The proof follows the same main idea in Section 5.1. The main difference from the synchronous case is that the weight matrix $W(k)$ is now random. As a result, the mixing-rate bound is stated in expectation, and the proof must also handle the random player-dependent stepsizes $\alpha_{k,i}$.

At each iteration k , only the pair (I^k, J^k) updates. We define

$$p_i = \frac{1}{N} \left(1 + \sum_{j \in N_i} p_{ji} \right),$$

which is the probability that player i updates at iteration k . The first term covers the case “ i wakes up”. The second term covers the case “a neighbor j wakes up and contacts

i ". Thus, players who are selected more often by their neighbors have larger p_i . We use $\mathcal{F}_k := \sigma(x^0, v^0, \{(I^l, J^l) : l < k\})$ to denote all gossip choices before iteration k . Then v^k and x^k are $\mathcal{F}_k$ measurable, while (I^k, J^k) is independent of $\mathcal{F}_k$.

5.2.1 Preliminary lemmas

Recall from Section 4.3 that the asynchronous stepsize is $\alpha_{k,i} = 1/\Gamma_i^k$. Since Γ_i^k is random, so is $\alpha_{k,i}$. For the convergence proof, we need bounds on the behavior of $\alpha_{k,i}$. The following stepsize properties are taken from [2, Lemma 7 and Corollary 1] and will be used in the convergence proof.

Lemma 5.7. Let Assumption 7 hold, and let $\alpha_{k,i} = 1/\Gamma_i^k$ for all i and k , where Γ_i^k denotes the number of updates of player i up to and including iteration k . Then, almost surely, the following hold for every $i \in \mathcal{V}$:

- (a) there exists a random index $m_i(\omega)$ such that $\alpha_{k,i} \leq \frac{2}{kp_i}$ for all $k \geq m_i(\omega)$;
- (b) $\sum_{k=1}^{\infty} \frac{\alpha_{k,i}}{k} < \infty$;
- (c) $\sum_{k=1}^{\infty} \alpha_{k,i}^2 < \infty$;
- (d) $\sum_{k=1}^{\infty} \left| \alpha_{k,i} - \frac{1}{kp_i} \right| < \infty$.

In addition, $\sum_{k=1}^{\infty} \alpha_{k,i} = \infty$ almost surely, since $\alpha_{k,i} \geq 1/k$ for all k , and hence the series dominates the harmonic series.

The synchronous mixing-rate bound (Lemma 5.3) used a deterministic transition matrix. In the asynchronous case, $W(k)$ is random, so the bound becomes a contraction in expectation. Define

$$D(k) = W(k) - \frac{1}{N} \mathbf{1}\mathbf{1}^\top, \quad (23)$$

the matrix $D(k)$ measures the deviation of $W(k)$ from the averaging matrix $\frac{1}{N} \mathbf{1}\mathbf{1}^\top$.

Lemma 5.8. Let Assumption 7 hold. Then there exists a $\lambda \in (0, 1)$, such that

$$\mathbb{E}[\|D(k)z\|^2 \mid \mathcal{F}_k] \leq \lambda \|z\|^2, \quad \forall z \in \mathbb{R}^N,$$

where $D(k)$ is defined in (23).

Proof. Since each $W(k)$ is doubly stochastic, we have

$$W(k) - \frac{1}{N} \mathbf{1}\mathbf{1}^\top W(k) = W(k) - \frac{1}{N} \mathbf{1}\mathbf{1}^\top = D(k).$$

By [2, Lemma 8], there exists $\lambda \in (0, 1)$ such that

$$\mathbb{E}[\|D(k)z\|^2] \leq \lambda \|z\|^2 \quad \text{for all } z \in \mathbb{R}^N.$$

In our setting, $W(k)$ is determined only by the current gossip pair (I^k, J^k) . Hence $D(k)$ also depends only on (I^k, J^k) . Moreover, by construction of the asynchronous gossip process, the pair (I^k, J^k) is independent of $\mathcal{F}_k$. Therefore, for every fixed $z \in \mathbb{R}^N$,

$$\mathbb{E}[\|D(k)z\|^2 \mid \mathcal{F}_k] = \mathbb{E}[\|D(k)z\|^2].$$

Combining the two relations gives

$$\mathbb{E}[\|D(k)z\|^2 \mid \mathcal{F}_k] \leq \lambda \|z\|^2, \quad \forall z \in \mathbb{R}^N,$$

which proves the claim. $\square$

Remark 5.1. The constant λ characterizes the mixing rate of the gossip step. A smaller value of λ means stronger contraction of the disagreement and hence faster consensus.

Since $v_i^k \in \mathbb{R}^n$ while Lemma 5.8 applies to vectors in $\mathbb{R}^N$, we argue coordinate-wise. For any vector $q \in \mathbb{R}^n$, let q_r denote its r -th component, where $r = 1, \dots, n$. For each fixed r define

$$v^k(r) = ([v_1^k]_r, \dots, [v_N^k]_r)^\top \in \mathbb{R}^N, \quad y^k(r) = [y^k]_r.$$

We then consider the disagreement vector

$$q^k(r) = v^k(r) - y^k(r)\mathbf{1}. \tag{24}$$

Lemma 5.9. Let Assumptions 1, 3, and 7 hold, and let C be the constant from Lemma 5.5 and $D(k)$ defined in (23). Then for every r and $k \geq 0$,

$$\|v^{k+1}(r) - y^{k+1}(r)\mathbf{1}\| \leq \|D(k)(v^k(r) - y^k(r)\mathbf{1})\| + \sqrt{2}C \sum_{i=1}^N \alpha_{k,i}.$$

Proof. We argue coordinate-wise and follow the proof idea of [2, Lemma 9], adapted to our notation. Fix $r \in \{1, \dots, n\}$ and define

$$\eta^{k+1}(r) = x^{k+1}(r) - x^k(r).$$

Since only two agents update at iteration k , the vector $\eta^{k+1}(r)$ has at most two nonzero

entries. From the mixing step (11) and the estimate update (13), we obtain

$$v^{k+1}(r) = W(k)v^k(r) + \eta^{k+1}(r).$$

To study disagreement, we subtract the consensus component $y^{k+1}(r)\mathbf{1}$. Using the definition of $y^{k+1}(r)$, the doubly stochastic property of $W(k)$, and the identity $D(k) = W(k) - \frac{1}{N}\mathbf{1}\mathbf{1}^\top$, we obtain

$$v^{k+1}(r) - y^{k+1}(r)\mathbf{1} = D(k)(v^k(r) - y^k(r)\mathbf{1}) + \left(\mathbf{I} - \frac{1}{N}\mathbf{1}\mathbf{1}^\top\right)\eta^{k+1}(r).$$

Since $\mathbf{I} - \frac{1}{N}\mathbf{1}\mathbf{1}^\top$ is an orthogonal projection, its operator norm is 1. Therefore,

$$\|v^{k+1}(r) - y^{k+1}(r)\mathbf{1}\| \leq \|D(k)(v^k(r) - y^k(r)\mathbf{1})\| + \|\eta^{k+1}(r)\|.$$

It remains to bound the update term. By the nonexpansiveness of Π_{X_i} (16) and Lemma 5.5, each nonzero entry of $\eta^{k+1}(r)$ is bounded by $\alpha_{k,i}C$. Since at most two entries are nonzero, we obtain

$$\|\eta^{k+1}(r)\| \leq \sqrt{2}C \sum_{i=1}^N \alpha_{k,i}.$$

Combining the two estimates proves the claim. $\square$

Lemma 5.9 bounds the disagreement for each coordinate. For the convergence proof, we need a stronger summability property.

Lemma 5.10. Let Assumptions 1–3 and 7 hold and let y^k be the estimated average defined in (17). Then for all $i \in \mathcal{V}$,

$$\sum_{k=1}^{\infty} \frac{1}{k} \|v_i^k - y^k\| < \infty \quad \text{a.s.}$$

Proof. The proof follows [2, Lemma 10] and combines three tools from above: the gossip contraction estimate (Lemma 5.8, originally due to [4, Sec. II-B]), the per-coordinate disagreement recursion (Lemma 5.9), and the supermartingale result (Lemma 5.1).

Consider the vector of the r -th coordinates of the players' estimates defined in equation (24). By Lemma 5.9,

$$\|q^{k+1}(r)\| \leq \|D(k)q^k(r)\| + \sqrt{2}C \sum_{i=1}^N \alpha_{k,i}.$$

Take conditional expectation, using Lemma 5.8 and Jensen's inequality,

$$\mathbb{E}[\|D(k)z\| \mid \mathcal{F}_k] \leq \sqrt{\mathbb{E}[\|D(k)z\|^2 \mid \mathcal{F}_k]} \leq \sqrt{\lambda} \|z\|$$

with $\lambda \in (0, 1)$. This gives the contraction

$$\mathbb{E}[\|q^{k+1}(r)\| \mid \mathcal{F}_k] \leq \sqrt{\lambda} \|q^k(r)\| + \sqrt{2}C \sum_i \alpha_{k,i} \quad \text{a.s.} \quad (25)$$

Now define

$$V_k = \frac{1}{k} \|q^k(r)\|, \quad k \geq 1.$$

Since

$$V_{k+1} = \frac{1}{k+1} \|q^{k+1}(r)\|,$$

we have

$$\mathbb{E}[V_{k+1} \mid \mathcal{F}_k] = \frac{1}{k+1} \mathbb{E}[\|q^{k+1}(r)\| \mid \mathcal{F}_k].$$

Dividing (25) by $k+1$ therefore gives

$$\mathbb{E}[V_{k+1} \mid \mathcal{F}_k] \leq \sqrt{\lambda} \frac{1}{k+1} \|q^k(r)\| + \frac{\sqrt{2}C}{k+1} \sum_{i=1}^N \alpha_{k,i}.$$

Since $\frac{k}{k+1} \leq 1$, we have

$$\sqrt{\lambda} \frac{1}{k+1} \|q^k(r)\| = \sqrt{\lambda} \frac{k}{k+1} V_k \leq \sqrt{\lambda} V_k.$$

Therefore,

$$\mathbb{E}[V_{k+1} \mid \mathcal{F}_k] \leq V_k - (1 - \sqrt{\lambda}) V_k + \frac{\sqrt{2}C}{k+1} \sum_{i=1}^N \alpha_{k,i}.$$

Let

$$R_k = \frac{1-\sqrt{\lambda}}{k} \|q^k(r)\|, \quad H_k = \frac{\sqrt{2}C}{k+1} \sum_{i=1}^N \alpha_{k,i}.$$

Then

$$\mathbb{E}[V_{k+1} \mid \mathcal{F}_k] \leq V_k - R_k + H_k.$$

It remains to show that $\sum_k H_k < \infty$ almost surely. By Lemma 5.7(c), we have

$$\sum_{k=1}^{\infty} \alpha_{k,i}^2 < \infty \quad \text{a.s. for all } i \in \mathcal{V}.$$

Since $\alpha_{k,i} = 1/\Gamma_k(i)$ and $\Gamma_k(i) \leq k+1$, we have

$$\frac{1}{k+1}\alpha_{k,i} \leq \alpha_{k,i}^2.$$

It follows that

$$\sum_{k=1}^{\infty} H_k \leq \sqrt{2} C \sum_{i=1}^N \sum_{k=1}^{\infty} \alpha_{k,i}^2 < \infty \quad \text{a.s.}$$

We may now apply Lemma 5.1. Hence

$$\sum_{k=1}^{\infty} R_k < \infty \quad \text{a.s.}$$

Since $R_k = (1 - \sqrt{\lambda})V_k$, this implies

$$\sum_{k=1}^{\infty} \frac{1}{k} \|q^k(r)\| < \infty \quad \text{a.s. for all } r.$$

Finally, fix $i \in \mathcal{V}$. Since the i -th component of $q^k(r)$ is $v_i^k(r) - y^k(r)$, we have

$$|v_i^k(r) - y^k(r)| = |[q^k(r)]_i| \leq \|q^k(r)\|.$$

Thus,

$$\|v_i^k - y^k\| \leq \sum_{r=1}^n |v_i^k(r) - y^k(r)| \leq \sum_{r=1}^n \|q^k(r)\|.$$

Hence,

$$\sum_{k=1}^{\infty} \frac{1}{k} \|v_i^k - y^k\| \leq \sum_{r=1}^n \sum_{k=1}^{\infty} \frac{1}{k} \|q^k(r)\| < \infty \quad \text{a.s.}$$

This proves the claim. $\square$

5.2.2 Asynchronous convergence theorem

We now prove almost sure convergence of the asynchronous gossip algorithm by using the preceding stepsize and disagreement estimates. We follow [2, Proposition 3].

Theorem 5.2. Let Assumptions 1–3 and 7 hold. Then the sequence $\{x^k\}$ generated by the gossip algorithm (11)–(13) with stepsize $\alpha_{k,i} = 1/\Gamma_i^k$ converges almost surely to the unique Nash equilibrium x^* of $\text{VI}(\mathcal{X}, \phi)$:

$$x^k \xrightarrow{\text{a.s.}} x^* \quad \text{as } k \rightarrow \infty.$$

Proof. By Proposition 3.1, the variational inequality $\text{VI}(\mathcal{X}, \phi)$ has a unique solution x^* . The proof follows the same idea as Theorem 5.1, only the selected pair $\{I^k, J^k\}$ updates,

and the stepsizes are random.

Let

$$V_k = \|x^k - x^*\|^2.$$

Step 1. Since x^* solves $\text{VI}(\mathcal{X}, \phi)$, the fixed-point formulation (3) gives

$$x_i^* = \Pi_{X_i}(x_i^* - \alpha_{k,i} F_i(x_i^*, \bar{x}^*))$$

for every $\alpha_{k,i} > 0$. For $i \in \{I^k, J^k\}$, using the nonexpansiveness of Π_{X_i} (16) and the update rule (12), we obtain:

$$\begin{aligned} \|x_i^{k+1} - x_i^*\|^2 &\leq \|x_i^k - x_i^*\|^2 + \alpha_{k,i}^2 \|F_i(x_i^k, N\hat{v}_i^k) - F_i(x_i^*, \bar{x}^*)\|^2 \\ &\quad - 2\alpha_{k,i} \left(F_i(x_i^k, N\hat{v}_i^k) - F_i(x_i^*, \bar{x}^*) \right)^\top (x_i^k - x_i^*). \end{aligned}$$

For $i \notin \{I^k, J^k\}$, we have $x_i^{k+1} = x_i^k$, and therefore $\|x_i^{k+1} - x_i^*\|^2 = \|x_i^k - x_i^*\|^2$.

Step 2. For $i \in \{I^k, J^k\}$, applying the same trick from Theorem 5.1, add and subtract $F_i(x_i^k, Ny^k)$ in the last term. Then

$$\begin{aligned} &\left(F_i(x_i^k, N\hat{v}_i^k) - F_i(x_i^*, \bar{x}^*) \right)^\top (x_i^k - x_i^*) \\ &= \left(F_i(x_i^k, Ny^k) - F_i(x_i^*, \bar{x}^*) \right)^\top (x_i^k - x_i^*) \\ &\quad + \left(F_i(x_i^k, N\hat{v}_i^k) - F_i(x_i^k, Ny^k) \right)^\top (x_i^k - x_i^*). \end{aligned}$$

Also write

$$\alpha_{k,i} = \left(\alpha_{k,i} - \frac{1}{kp_i} \right) + \frac{1}{kp_i}.$$

By the same boundedness and Lipschitz arguments as in Theorem 5.1, there exist constants $C_1, C_2, C_3 > 0$, independent of k , such that

$$\|F_i(x_i^k, N\hat{v}_i^k) - F_i(x_i^*, \bar{x}^*)\|^2 \leq C_1$$

and

$$\left| \left(F_i(x_i^k, Ny^k) - F_i(x_i^*, \bar{x}^*) \right)^\top (x_i^k - x_i^*) \right| \leq C_2.$$

Moreover, by Cauchy–Schwarz and Assumption 3,

$$\begin{aligned} &\left(F_i(x_i^k, N\hat{v}_i^k) - F_i(x_i^k, Ny^k) \right)^\top (x_i^k - x_i^*) \\ &\geq -\|F_i(x_i^k, N\hat{v}_i^k) - F_i(x_i^k, Ny^k)\| \|x_i^k - x_i^*\| \\ &\geq -C_3 \|\hat{v}_i^k - y^k\|. \end{aligned}$$

Combining these estimates with step 1 gives

$$\begin{aligned} \|x_i^{k+1} - x_i^*\|^2 &\leq \|x_i^k - x_i^*\|^2 + C_1 \alpha_{k,i}^2 + C_2 \left| \alpha_{k,i} - \frac{1}{kp_i} \right| \\ &\quad - \frac{2}{kp_i} \left(F_i(x_i^k, Ny^k) - F_i(x_i^*, \bar{x}^*) \right)^\top (x_i^k - x_i^*) + C_3 \alpha_{k,i} \|\hat{v}_i^k - y^k\|. \end{aligned} \quad (26)$$

Finally, since

$$\hat{v}_i^k = \sum_{j=1}^N W_{ij}^k v_j^k,$$

convexity of the norm (14) gives

$$\|\hat{v}_i^k - y^k\| \leq \sum_{j=1}^N W_{ij}^k \|v_j^k - y^k\|.$$

Hence, after summing over i and using that W^k is doubly stochastic, the total disagreement term is bounded by a constant multiple of $\sum_{i=1}^N \alpha_{k,i} \|v_i^k - y^k\|$.

Step 3. We now take conditional expectation with respect to $\mathcal{F}_k$. For a fixed player i , the estimate in step 2 applies only on the event $i \in \{I^k, J^k\}$, which has probability p_i . Since x^k and v^k are $\mathcal{F}_k$ -measurable, we obtain

$$\begin{aligned} \mathbb{E}[\|x_i^{k+1} - x_i^*\|^2 \mid \mathcal{F}_k] &\leq \|x_i^k - x_i^*\|^2 + C_1 \mathbb{E}[\alpha_{k,i}^2 \mid \mathcal{F}_k] + C_2 \mathbb{E} \left[\left| \alpha_{k,i} - \frac{1}{kp_i} \right| \mid \mathcal{F}_k \right] \\ &\quad - 2p_i \frac{1}{kp_i} \left(F_i(x_i^k, Ny^k) - F_i(x_i^*, \bar{x}^*) \right)^\top (x_i^k - x_i^*) \\ &\quad + C_3 \mathbb{E}[\alpha_{k,i} \|v_i^k - y^k\| \mid \mathcal{F}_k]. \end{aligned}$$

Hence

$$\begin{aligned} \mathbb{E}[\|x_i^{k+1} - x_i^*\|^2 \mid \mathcal{F}_k] &\leq \|x_i^k - x_i^*\|^2 + C_1 \mathbb{E}[\alpha_{k,i}^2 \mid \mathcal{F}_k] + C_2 \mathbb{E} \left[\left| \alpha_{k,i} - \frac{1}{kp_i} \right| \mid \mathcal{F}_k \right] \\ &\quad - \frac{2}{k} \left(F_i(x_i^k, Ny^k) - F_i(x_i^*, \bar{x}^*) \right)^\top (x_i^k - x_i^*) \\ &\quad + C_3 \mathbb{E}[\alpha_{k,i} \|v_i^k - y^k\| \mid \mathcal{F}_k]. \end{aligned}$$

Summing over $i = 1, \dots, N$ and using $Ny^k = \sum_{j=1}^N x_j^k$ from Lemma 5.4, we obtain

$$\mathbb{E}[V_{k+1} \mid \mathcal{F}_k] \leq V_k - R_k + H_k \quad \text{a.s.,}$$

where

$$R_k = \frac{2}{k} (\phi(x^k) - \phi(x^*))^\top (x^k - x^*) \geq 0,$$

and H_k is bounded by a finite linear combination of

$$\sum_{i=1}^N \mathbb{E}[\alpha_{k,i}^2 \mid \mathcal{F}_k], \quad \sum_{i=1}^N \mathbb{E}\left[\left|\alpha_{k,i} - \frac{1}{kp_i}\right| \mid \mathcal{F}_k\right], \quad \sum_{i=1}^N \frac{1}{k} \|v_i^k - y^k\|.$$

Step 4. By Lemma 5.7,

$$\sum_{k=1}^{\infty} \alpha_{k,i}^2 < \infty, \quad \sum_{k=1}^{\infty} \left|\alpha_{k,i} - \frac{1}{kp_i}\right| < \infty,$$

and by Lemma 5.10,

$$\sum_{k=1}^{\infty} \frac{1}{k} \|v_i^k - y^k\| < \infty \quad \text{a.s.}$$

for every $i \in \mathcal{N}$. Therefore,

$$\sum_{k=1}^{\infty} H_k < \infty \quad \text{a.s.}$$

Applying Lemma 5.1 with $a_k = 0$, we obtain that, almost surely, V_k converges, and

$$\sum_{k=1}^{\infty} \frac{2}{k} \left(\phi(x^k) - \phi(x^*) \right)^\top (x^k - x^*) < \infty \quad \text{a.s.}$$

The same subsequence argument as in Theorem 5.1, together with the strict monotonicity of ϕ , implies that the only possible accumulation point of $\{x^k\}$ is x^* . Since $V_k = \|x^k - x^*\|^2$ converges almost surely and any convergent subsequence of $\{x^k\}$ must converge to x^* , the limit of V_k must be zero. Hence, $x^k \rightarrow x^*$ a.s. $\square$

6 Numerical experiments

This chapter runs the two distributed algorithms from Chapter 4 on several communication graphs. We follow the setup in [2] and state where we differ. We consider four baseline graphs: Cycle, Wheel, Grid, and Complete. We then extend the study to Watts–Strogatz small-world networks [3]. These networks combine high clustering with a small characteristic path length and are often used as models of real communication networks. We consider two extensions. First, we test both algorithms on one representative small-world graph. Second, we study how the rewiring probability p_{rew} affects convergence.

The source code is publicly available in the author’s GitHub repository [16]. The numerical study answers the following three research questions.

- Q1.** How closely does our implementation reproduce the benchmark results? We also test the synchronous algorithm on the same four baseline graphs used for the asynchronous benchmark in that paper.
- Q2.** How do the synchronous and asynchronous algorithms compare on a representative small-world communication network?
- Q3.** How does the rewiring probability p_{rew} in the Watts–Strogatz model affect the convergence of the synchronous and asynchronous distributed algorithms?

6.1 Setup

Recall the Nash–Cournot model in Chapter 3. Each firm $i \in \{1, \dots, N\}$ produces and sells in $\mathcal{L}$ markets. Let $g_i = (g_{i1}, \dots, g_{i\mathcal{L}}) \in \mathbb{R}^{\mathcal{L}}$ and $s_i = (s_{i1}, \dots, s_{i\mathcal{L}}) \in \mathbb{R}^{\mathcal{L}}$ be the production and sales vectors of firm i . Each firm i solves:

$$\begin{aligned} \min_{(g_i, s_i)} \quad & \sum_{\ell=1}^{\mathcal{L}} \left[c_{i\ell}(g_{i\ell}) - p_{\ell}(\bar{s}_{\ell}) s_{i\ell} \right] \\ \text{subject to} \quad & s_{i\ell}, g_{i\ell} \geq 0, \quad g_{i\ell} \leq \text{Cap}_{i\ell}, \quad \ell = 1, \dots, \mathcal{L}, \\ & \sum_{\ell=1}^{\mathcal{L}} s_{i\ell} = \sum_{\ell=1}^{\mathcal{L}} g_{i\ell}, \end{aligned}$$

In this model, firms are coupled only through the aggregate sales vector $\bar{s}$. More precisely, the price at market ℓ depends on aggregate sales $\bar{s}_{\ell} = \sum_i s_{i\ell}$. Although the feasible set in Chapter 3 is written as

$$\sum_{\ell=1}^{\mathcal{L}} s_{i\ell} \leq \sum_{\ell=1}^{\mathcal{L}} g_{i\ell},$$

in the numerical experiments we impose the equality constraint. Since production costs are strictly increasing and unsold output has zero value, no optimal solution involves excess production. We use inverse demand and quadratic production cost:

$$p_\ell(\bar{s}_\ell) = m_\ell - \bar{s}_\ell, \quad c_{i\ell}(g_{i\ell}) = a_{i\ell} g_{i\ell} + b_{i\ell} g_{i\ell}^2.$$

We fix $\mathcal{L} = 10$ and consider $N \in \{20, 50\}$ firms. The coefficients are drawn independently from uniform distributions on the following intervals:

$$m_\ell \sim \mathcal{U}[90, 100], \quad a_{i\ell} \sim \mathcal{U}[2, 12], \quad b_{i\ell} \sim \mathcal{U}[2, 3],$$

and we set $\text{Cap}_{i\ell} = 500$. For each network size N , the coefficients are sampled once and kept fixed across all graphs, runs, and checkpoints. This avoids differences caused by coefficient resampling. Under this specification, the game has a unique Nash equilibrium, see Proposition 3.2.

Communication graph and weight matrix. The baseline graphs (Cycle, Wheel, Grid, Complete) and the Watts–Strogatz graph family are defined in Chapter 2. For each graph $\mathcal{G}$, we define the doubly stochastic weight by

$$[W]_{ij} = \delta \text{ if } (i, j) \in \mathcal{E}, \quad [W]_{ii} = 1 - \delta d_i, \quad \delta = \frac{0.5}{\max_i d_i},$$

where d_i is the degree of node i . By construction, W is symmetric and doubly stochastic.

Sample paths, error metric, and initial point. Each player's joint decision is $x_i = (g_i, s_i) \in \mathbb{R}^{2\mathcal{L}}$, where g_i is the production vector and s_i the sales vector. For every table cell, we report results from $R = 50$ runs. Across runs, the variation comes from the graph realization, the initial point, and in the gossip algorithms, the gossip sequence.

We use the relative error from [2, eq. (65)]

$$\text{error}_{\tilde{k}} = \frac{\max_{i,\ell} \{ |g_{i\ell}^{\tilde{k}} - g_{i\ell}^*|, |s_{i\ell}^{\tilde{k}} - s_{i\ell}^*| \}}{\max_{i,\ell} \{ |g_{i\ell}^*|, |s_{i\ell}^*| \}}, \quad (27)$$

where (g^*, s^*) is the reference Nash equilibrium, computed by a centralized projected-gradient method using the true aggregate $\bar{s}$. The symbol $\tilde{k}$ denotes the fixed iteration horizon at which the relative error is evaluated.

We report the mean relative error over $R = 50$ runs and the full width of the two-sided

90 % normal-approximation confidence interval for the sample mean:

$$2z_{0.95} \frac{s_R}{\sqrt{R}}, \quad z_{0.95} = 1.645.$$

For the initial point, we draw $\xi_i \sim \mathcal{U}([0, 10])^{2\mathcal{L}}$ and set $x_i^0 = \Pi_{X_i}(\xi_i)$, so that the initialization is random and feasible.

Algorithms and stepsize rules. The two algorithms use different diminishing stepsizes. In the synchronous algorithm, we set $\alpha_k = 1/k$. In the asynchronous algorithm, we set $\alpha_{k,i} = 9/\Gamma_i^k$, where Γ_i^k is the number of updates of player i up to iteration k . For the asynchronous algorithm we use $p_{ij} = 1/d_i$, that is, conditional on player i being active, it selects a neighbor uniformly at random. This is the standard choice in [4]. The convergence Theorem 5.2 only requires $p_{ij} > 0$, so this is a valid choice. Note that experiments use a tuned constant factor 9 for numerical reproduction.

Spectral indicators. For the synchronous algorithm, we report

$$\rho_{\text{sync}} = \rho\left(W - \frac{1}{N}\mathbf{1}\mathbf{1}^\top\right),$$

the largest nontrivial eigenvalue magnitude of the fixed symmetric averaging matrix W . A smaller value indicates faster long-run mixing in the consensus step [17]. For the asynchronous gossip algorithm, letting $W(k)$ denote the random mixing matrix of one gossip update as in (11), we report

$$\rho_{\text{async}} = \sqrt{\lambda_2(\mathbb{E}[W])},$$

following [2]. These quantities are used only as spectral indicators of information mixing.

6.2 Validation and extension

This section answers **Q1**. We first check whether our implementation reproduces the benchmark results in [2, Sec. 6]. We then extend the study to the synchronous algorithm by running it on the same four baseline graphs: Cycle, Wheel, Grid and Complete.

Table 1: Centralized synchronous baseline on the static complete graph: mean error (27) and width of 90 % confidence interval after $\tilde{k}$ iterations.

N	$\tilde{k}$	Mean error	90 % CI width
20	5 000	8.48e-05	1.38e-05
20	10 000	4.24e-05	6.89e-06
50	5 000	2.37e-04	≈ 0
50	10 000	1.19e-04	≈ 0

Table 1 is a sanity check for the implementation. It also shows the error level when the aggregate is exact. This table serves as a reference point before we turn the distributed experiments on the four baseline graphs.

Table 2: Synchronous algorithm: mean error after $\tilde{k}$ iterations.

N	$\tilde{k}$	Cycle	Wheel	Grid	Complete
20	5 000	2.58e-02	3.47e-02	3.06e-02	9.98e-03
20	10 000	1.33e-02	1.85e-02	1.55e-02	5.00e-03
50	5 000	3.38e-01	2.55e-01	7.50e-01	3.12e-01
50	10 000	1.94e-01	1.41e-01	4.03e-01	1.57e-01

Table 3: Synchronous algorithm: width of 90 % confidence interval of the mean error after $\tilde{k}$ iterations.

N	$\tilde{k}$	Cycle	Wheel	Grid	Complete
20	5 000	2.44e-03	2.32e-03	2.75e-03	6.70e-04
20	10 000	1.27e-03	1.24e-03	1.41e-03	3.36e-04
50	5 000	5.11e-02	4.51e-02	1.11e-01	2.16e-02
50	10 000	3.04e-02	1.82e-02	6.01e-02	1.09e-02

Table 4: Asynchronous algorithm: mean error after gossip events.

N	$\tilde{k}$	Cycle	Wheel	Grid	Complete
20	$5 \cdot 10^4$	2.92e-02	1.59e-08	4.64e-04	4.08e-10
20	$1 \cdot 10^5$	1.39e-03	4.08e-10	5.67e-06	4.08e-10
50	$5 \cdot 10^4$	39.2	2.94e-02	3.18	3.08e-04
50	$1 \cdot 10^5$	6.14	2.45e-04	7.78e-01	1.33e-06

Table 5: Asynchronous algorithm: width of 90 % confidence interval of the mean error after gossip events.

N	$\tilde{k}$	Cycle	Wheel	Grid	Complete
20	$5 \cdot 10^4$	6.03e-04	2.09e-09	1.38e-05	2.17e-14
20	$1 \cdot 10^5$	3.21e-05	2.05e-14	1.83e-07	1.98e-14
50	$5 \cdot 10^4$	4.25	3.01e-03	1.80e-01	2.80e-05
50	$1 \cdot 10^5$	4.05e-01	2.65e-05	2.46e-02	1.22e-07

Table 2 and Table 3 report the synchronous results (extension), and Table 4 and Table 5 report the asynchronous results (validation) on the same graphs and the same values of N .

This comparison should be read with care, because synchronous rounds and gossip events are not the same budget unit. At $N = 20$, the reported horizons correspond to roughly the same per-player work, but at $N = 50$ the synchronous method performs about 2.5 times more per-player updates. This does not mean that a larger asynchronous stepsize is harmful by itself. Rather, on the Cycle and the Grid, aggregate information spreads slowly through the network. As a result, local estimates remain inaccurate for longer and the asynchronous updates become more sensitive to this consensus-tracking error. Thus, Table 2–Table 5 should be read as a comparison under benchmark conventions, not as a strictly equal-cost one.

Table 6: Number of gossip events $\tilde{k}_\epsilon$ required for the players’ aggregate estimates to agree within $\epsilon = 10^{-3}$. Reported are the sample mean and sample standard deviation over $R = 50$ runs.

Network	$\sqrt{\lambda_2(\mathbb{E}[W])}$	mean($\tilde{k}_\epsilon$)	std ($\tilde{k}_\epsilon$)
Cycle	0.9988	134 968	1 090
Wheel	0.9868	14 032	596
Grid	0.9971	55 916	799
Complete	0.9733	7 196	278

Table 6 provides additional evidence for the graph ordering in the asynchronous benchmark at $N = 20$. It reports the stopping time experiment based on the agreement threshold for the aggregate estimates. The table shows that the asynchronous method reaches the target much faster on the Complete and Wheel graphs than on the Cycle and Grid graphs.

Overall, the implementation reproduces the benchmark behavior well. The main effect

is that the asynchronous method has a clear advantage on well-mixed graphs, while the synchronous method becomes more competitive on slowly mixing graphs at $N = 50$. This supports the view that the communication Network is an important factor in the relative performance of the two methods.

6.3 Convergence on a representative small-world setting

We now compare the two algorithms on one representative Watts–Strogatz small-world network [3]. This comparison is meant to illustrate, not to rank. We first use graph statistics at several values of p_{rew} to choose one representative small-world setting. We then fix $p_{\text{rew}} = 0.1$ and average over $R = 50$ independent connected graphs from $WS(N = 50, K = 6, p_{\text{rew}} = 0.1)$. Both algorithms run on the same graphs and from the same initial points. Any difference therefore comes from the update rule, not from the sampled graph or the starting point.

6.3.1 Structural indicators across different values of p_{rew}

Table 7 reports two structural indicators of the graph: the clustering coefficient $C(\mathcal{G})$ and the characteristic path length $L(\mathcal{G})$. All values are sample means over $R = 50$ random graphs.

Table 7: Watts–Strogatz ($N = 50$, $K = 6$): clustering coefficient $C(\mathcal{G})$ and characteristic path length $L(\mathcal{G})$ for different values of the rewiring probability p_{rew} .

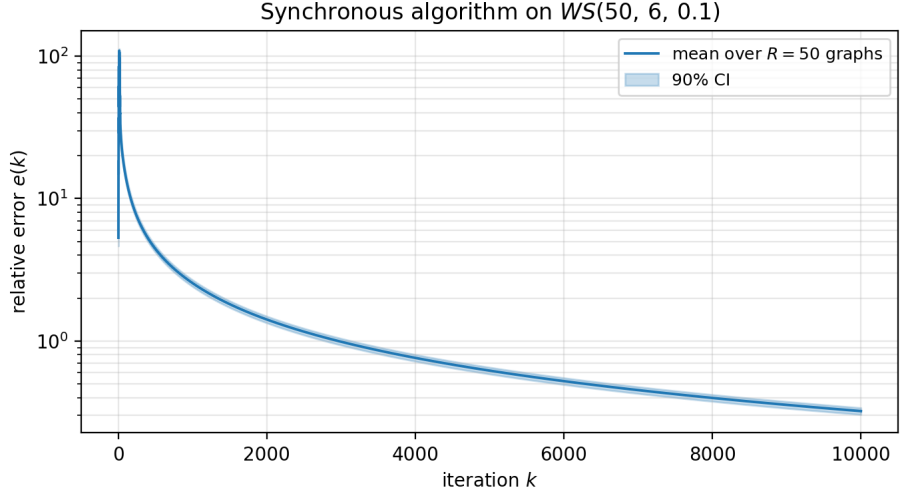
p_{rew}	0	0.001	0.005	0.01	0.05	0.1	0.5	1.0
$C(\mathcal{G})$	0.600	0.598	0.595	0.582	0.521	0.450	0.153	0.103
$L(\mathcal{G})$	4.59	4.54	4.38	4.08	3.24	2.87	2.36	2.31

Watts and Strogatz [3] describe two main features of small-world Networks. First, $L(\mathcal{G})$ drops rapidly after a small amount of rewiring. Second, $C(\mathcal{G})$ remains close to the lattice value over a range of p_{rew} . Table 7 shows that $p_{\text{rew}} = 0.1$ exhibits both features. $L(\mathcal{G})$ is already much smaller than at $p_{\text{rew}} = 0$ and $p_{\text{rew}} = 0.01$, and is close to its random-graph level. At the same time, the clustering coefficient remains well above the random case. For this reason, we use $p_{\text{rew}} = 0.1$ as the representative small-world setting in the comparison below.

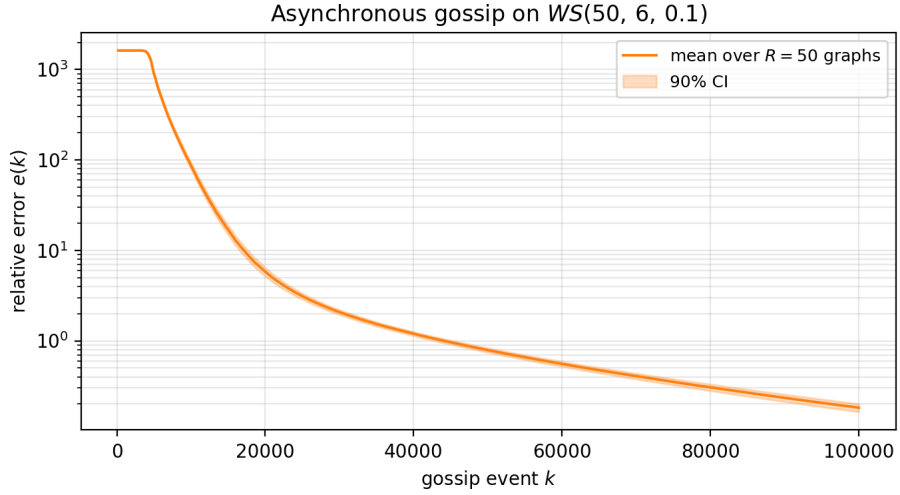
6.3.2 Convergence performance

The structural indicators above point to $p_{\text{rew}} = 0.1$. We now fix this value and average over $R = 50$ independent connected graphs of $WS(50, 6, 0.1)$. The two algorithms are

run on the same set of graph instances and from the same initial points. This gives a paired comparison. Figure 3a and Figure 3b show the convergence behavior of the synchronous and asynchronous algorithms on this representative setting.



(a) Synchronous algorithm



(b) Asynchronous gossip

Figure 3: Convergence on $WS(50, 6, 0.1)$. The k axis counts synchronous rounds in (a) and gossip events in (b). These axes show progress, not equal-cost budgets.

Table 8 reports the mean error at the horizons $\tilde{k} = 10^4$ for the synchronous algorithm and $\tilde{k} = 10^5$ for the asynchronous algorithm. This follows the reporting rule of Section 6.2. The Watts–Strogatz row can therefore be read next to the baseline tables.

Table 8: Watts–Strogatz (50, 6, 0.1): mean error and width of 90 % CI at the horizons $\tilde{k}$. Both algorithms are run on the same graph instances and from the same initial points.

Algorithm	$\tilde{k}$	Mean error	90 % CI width	ρ
Sync	10^4	0.322	3.53e-02	0.968
Async	10^5	0.182	3.76e-02	0.999

At the reported horizons, the asynchronous method attains a lower mean error than the synchronous method on the representative small-world setting. Both curves are still decreasing, so this should be read as a horizon-based comparison, not as a final equal-cost ranking. This is consistent with the fact that $WS(50, 6, 0.1)$ lies between the slowly mixing baseline graphs and the well-mixed ones.

6.4 Sensitivity to the rewiring probability p_{rew}

Section 6.3 fixed one value, $p_{\text{rew}} = 0.1$. We now vary it and ask how the rewiring probability affects convergence. This answers **Q3**.

We fix $N = 50$ and $K = 6$, and test

$$p_{\text{rew}} \in \{0, 0.001, 0.005, 0.01, 0.05, 0.1, 0.5, 1.0\}.$$

As p_{rew} increases, the Watts–Strogatz graph changes from a ring lattice to a more random graph.

For each p_{rew} , we generate $R = 50$ independent connected graphs $WS(50, 6, p_{\text{rew}})$ and run both algorithms on them. The two algorithms use the same $R = 50$ graphs and the same initial points. Any difference therefore comes from the update rule, not from the graph or the starting point. These $R = 50$ graphs are sampled independently from those used in Section 6.3. Hence the $WS(50, 6, 0.1)$ values in Table 9 differ slightly from Table 8.

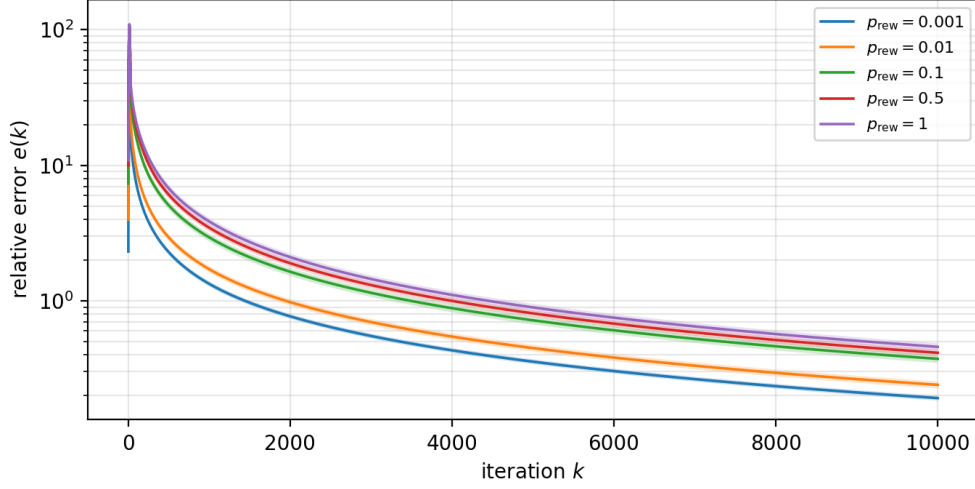
The synchronous algorithm stops at $\tilde{k} = 10^4$, and the asynchronous algorithm stops at $\tilde{k} = 10^5$. These are the same horizons as in Section 6.2.

Figure 4a and Figure 4b show the mean relative error for five values of p :

$$p_{\text{rew}} \in \{0.001, 0.01, 0.1, 0.5, 1.0\}.$$

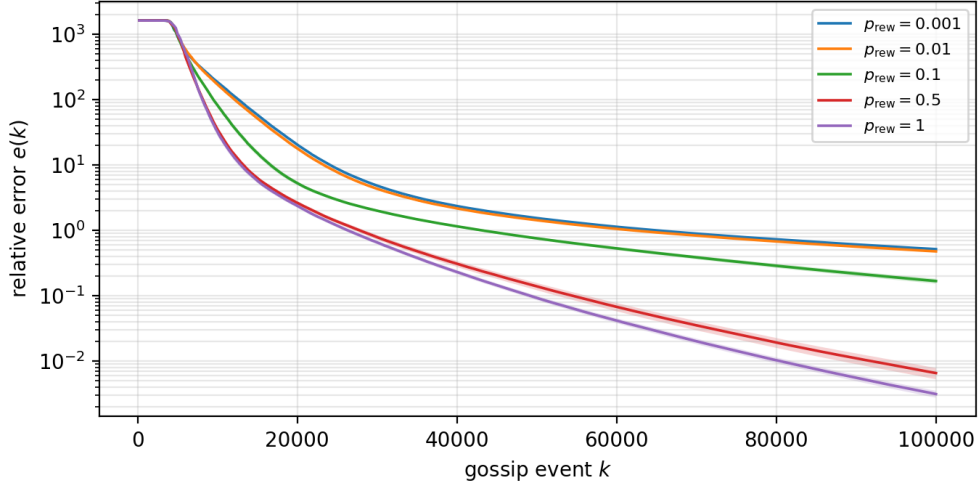
We plot only these five values to keep the figures readable. Table 9 reports the full eight-point sweep, including $p_{\text{rew}} = 0$.

Sync sensitivity to rewiring probability p_{rew} , $WS(50, 6, p_{\text{rew}})$



(a) Synchronous algorithm: mean error against iteration k .

Async sensitivity to rewiring probability p_{rew} , $WS(50, 6, p_{\text{rew}})$



(b) Asynchronous gossip: mean relative error against gossip event k .

Figure 4: Convergence on $WS(50, 6, p_{\text{rew}})$ for five values of the rewiring probability p_{rew} . Both algorithms are run on the same $R = 50$ graphs and from the same initial points (paired comparison).

Table 9 shows a clear sensitivity difference. As p_{rew} increases, the asynchronous mean error decreases strongly, while the synchronous mean error increases. This suggests that added rewiring is much more helpful for the asynchronous gossip algorithm. For the asynchronous method, the change in the mean error is much larger than the change in ρ_{async} . A more natural explanation is that shorter paths help aggregate information spread more quickly through the networks.

Table 9: $WS(50, 6, p_{\text{rew}})$: sensitivity to the rewiring probability p_{rew} . The synchronous and asynchronous algorithms use the same set of graphs (paired comparison).

p_{rew}	Sync ($\tilde{k} = 10^4$)		Async ($\tilde{k} = 10^5$)		Spectral indicator ρ	
	mean	90 % CI	mean	90 % CI	sync	async
0	0.171	1.34e-02	0.514	1.05e-02	0.9817	0.9996
0.001	0.192	1.64e-02	0.514	9.24e-03	0.9821	0.9996
0.005	0.210	2.54e-02	0.496	1.77e-02	0.9830	0.9996
0.01	0.240	2.85e-02	0.475	2.43e-02	0.9828	0.9996
0.05	0.296	4.43e-02	0.307	3.66e-02	0.9764	0.9994
0.1	0.374	5.05e-02	0.167	2.92e-02	0.9677	0.9992
0.5	0.414	4.95e-02	0.007	2.55e-03	0.9197	0.9974
1.0	0.458	5.76e-02	0.003	7.51e-04	0.9145	0.9970

We now relate these results to the mixing constants from the convergence analysis. For the synchronous algorithm, the disagreement decays at the geometric rate β from Lemma 5.3. We summarize this rate by ρ_{sync} . For the asynchronous algorithm, Lemma 5.8 gives a one-step contraction λ . We summarize it by $\rho_{\text{async}} = \sqrt{\lambda_2(\mathbb{E}[W])}$. A smaller value means faster mixing.

As p_{rew} increases, the path length L drops from about 4.59 to 2.31. At the same time, ρ_{async} stays almost constant (from 0.9996 to 0.9970), see Table 7. However, the asynchronous mean error drops by more than two orders of magnitude. So ρ_{async} alone cannot explain this. The shorter path length is a more direct cause. Shorter paths let the aggregate information reach all players in fewer gossip steps.

The synchronous case is different. Here ρ_{sync} decreases (from 0.982 to 0.915). This suggests faster mixing, but the mean error at the fixed horizon still grows. The indicator ρ only describes the long-run per-step rate. It does not fully explain the behavior at a fixed horizon. So we do not claim a single cause for the synchronous case.

6.5 Main findings

Section 6.2–Section 6.4 show clear differences between the two algorithms. The general comparison in Section 6.2 and the paired experiments in Section 6.3 and Section 6.4 all point in the same direction.

The communication graph structure matters for both algorithms. Even on the same graph instances and the same initial points, the two methods do not behave in the same way.

The rewiring probability affects the two methods differently. As p_{rew} increases, the asynchronous method improves strongly, while the synchronous method does not benefit and its error even grows slightly at the reported horizon. The spectral indicator ρ does not fully explain this. A more plausible explanation is that shorter paths allow aggregate information to spread more quickly across the network.

Overall, the experiments indicate that the network structure strongly affects finite horizon convergence. They also show that the two algorithms react differently to the same change in graph structure.

7 Conclusion and limitations

This thesis studied two distributed algorithms for aggregative games on communication networks: a synchronous algorithm and an asynchronous gossip algorithm. In the numerical experiments, four baseline graphs were considered, with particular attention to Watts–Strogatz small-world graphs. The theoretical framework from [2] was adapted to the present setting, and the numerical study was carried out for the two distributed algorithms.

The main contribution of this thesis is empirical. For Q1, the benchmark results were largely reproduced, and the study was extended to the synchronous algorithm on the same baseline graphs. For Q2, on a representative small-world graph, the asynchronous algorithm achieved a lower mean error at the reported horizons. For Q3, the rewiring probability p_{rew} affected the two algorithms differently: as p_{rew} increased, the asynchronous algorithm improved, while the synchronous algorithm became less accurate. For the asynchronous algorithm, this improvement is explained better by the shorter communication paths than by the spectral indicator alone. Overall, the results show that the communication graph has a strong effect on convergence.

This thesis also has some limitations. The comparison is based on fixed reporting horizons, and synchronous rounds and gossip events are not the same budget unit. So the results should not be read as a final equal-cost ranking. Only one game model and relatively limited network sizes were considered. In addition, for the asynchronous algorithm, a diminishing stepsize with constant factor 9 was used without further analysis of this choice. A natural next step is to test other values of this constant on the same graphs. Another direction is to study other graph models, such as scale-free networks [18].

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Declaration of Authorship

I declare that this thesis is my own work, that all statements and information contained herein are true and accurate to the best of my knowledge, and that no sources or aids have been used other than those acknowledged. During the preparation of this thesis, I used AI tools (Anthropic's Claude) to assist with writing and debugging the Python code for the numerical experiments. I reviewed, tested, and verified all generated code, and I take full responsibility for the results presented. This thesis has not been previously submitted, in whole or in part, for assessment at any other institution.

Mannheim, 22.06.2026

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